



ENCLOSURE

Intro to CAD Guide:

Design a custom enclosure for your project, get a 3D print of it shipped to you, and win other prizes!



Fusion 360: Table of Contents

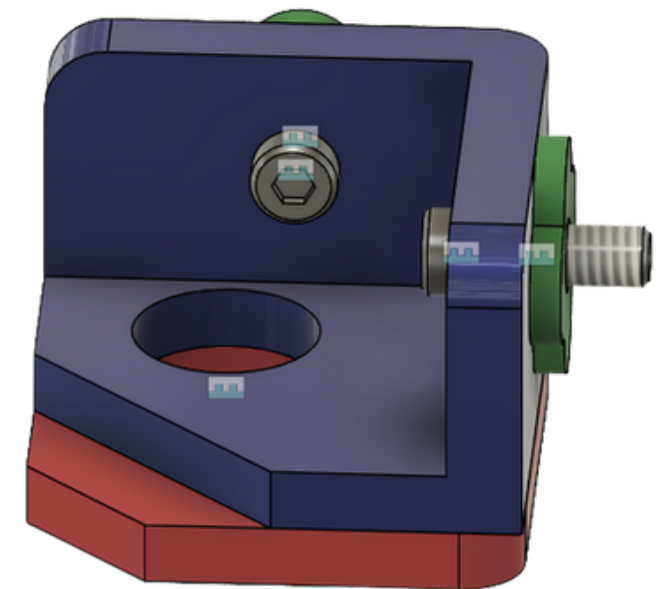
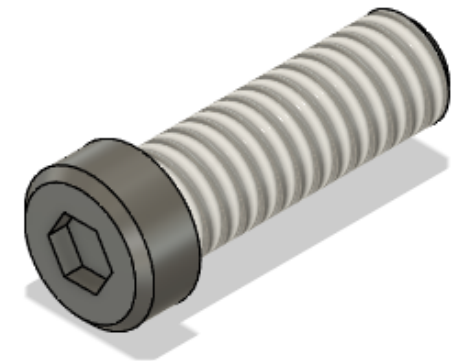
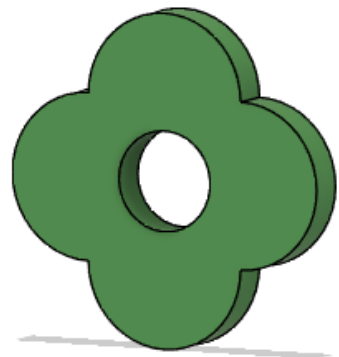
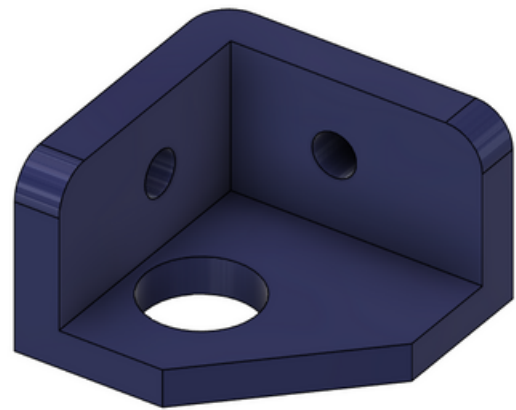
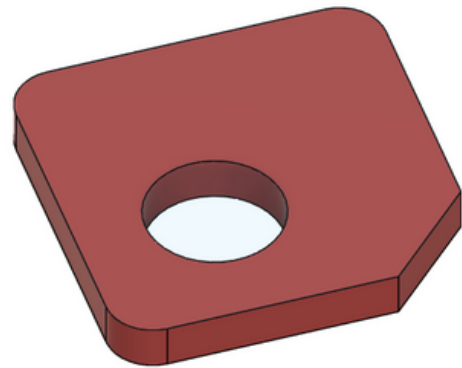
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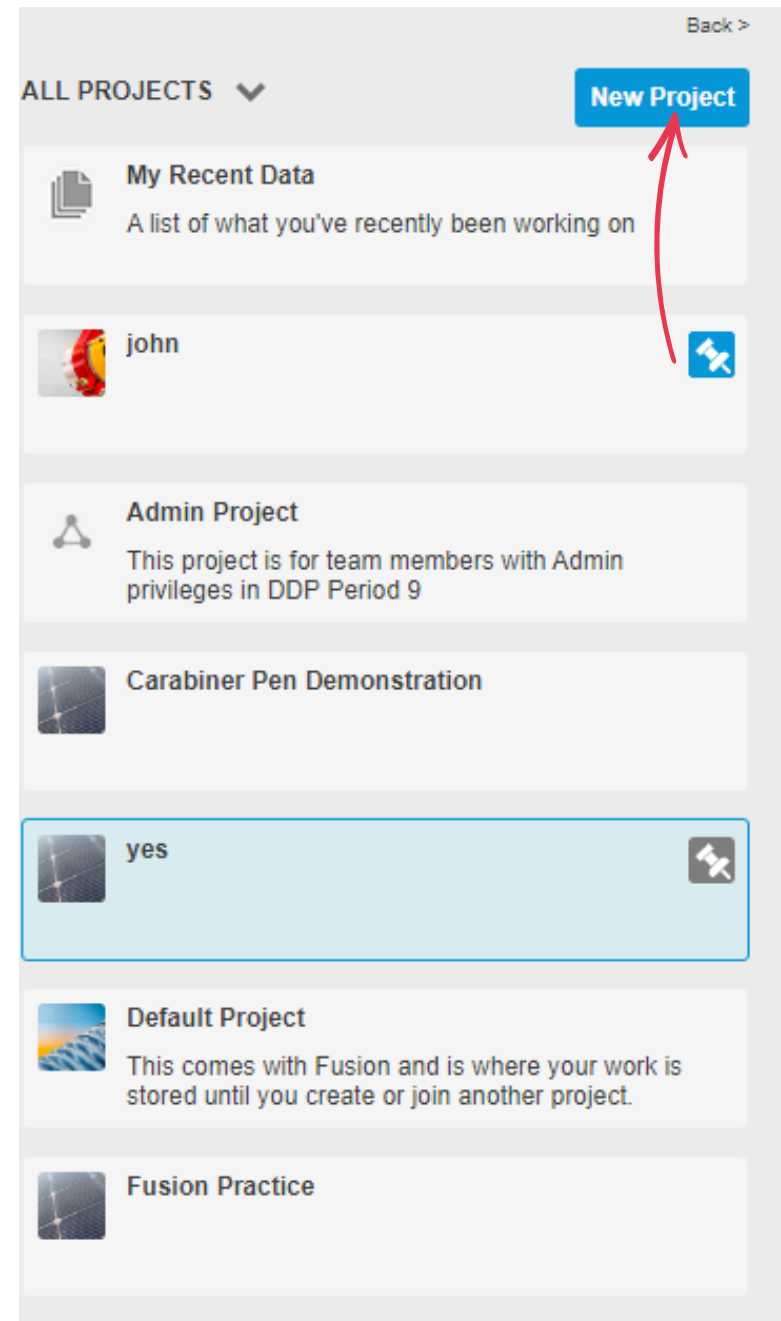
Objectives

1. **Fusion Basics:** Learn core tools and create simple sketches.
2. **3D Print Prep:** Properly dimension models for 3D printing.
3. **Hands-on Practice:** Follow steps to build three basic models.
4. **Assembly & Joints:** Use the joint command for functional and multi-part designs.
5. **Custom Design:** Apply skills to design a unique CAD enclosure from scratch!

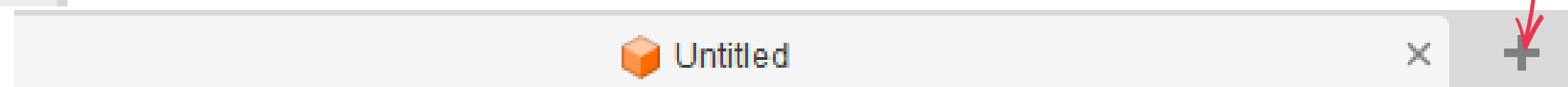




Getting Started



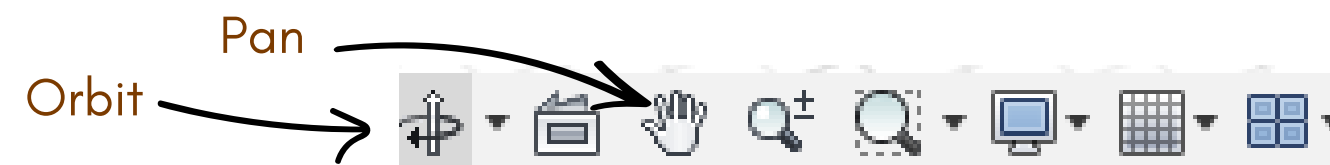
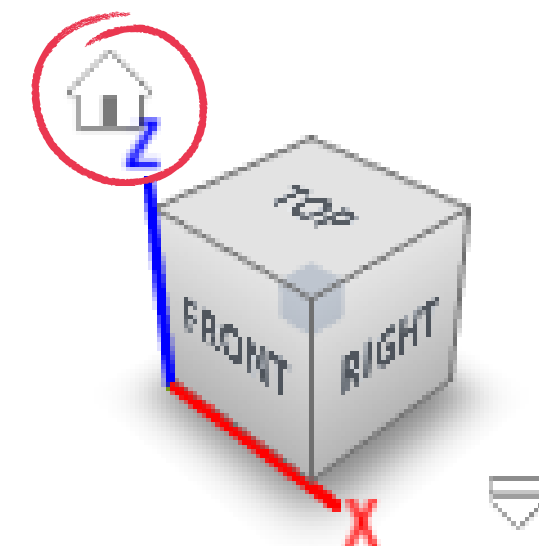
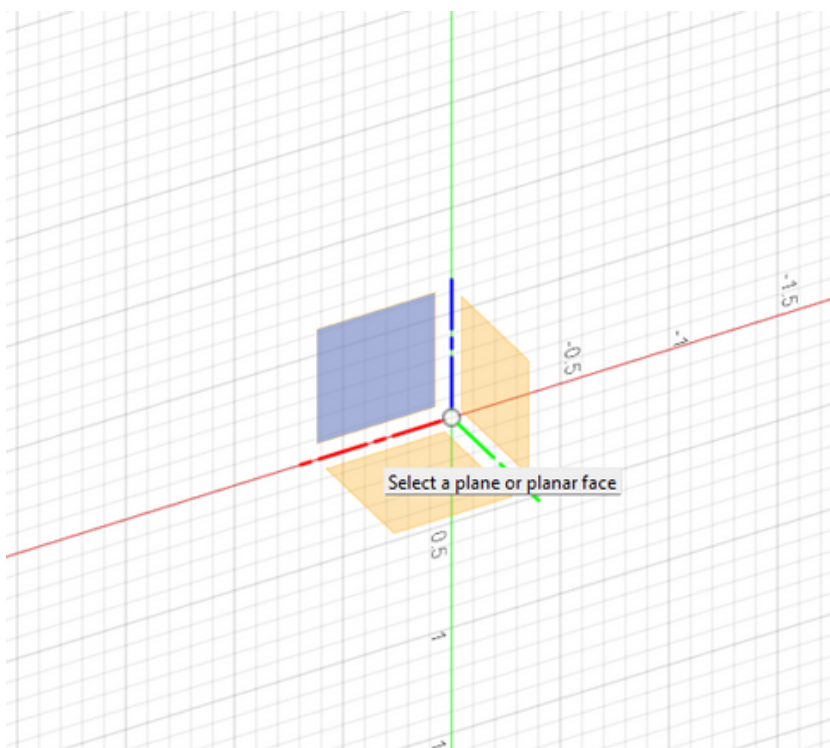
Welcome to Fusion 360! Let's get started by making a new project. You may need to start a new design to create a project. To create a new design, click the plus at the top of your screen. Next, select the "new project" button on the left side of your screen and name it "Introduction to Fusion". We can now dive into the basics of Fusion 360.





Camera Settings

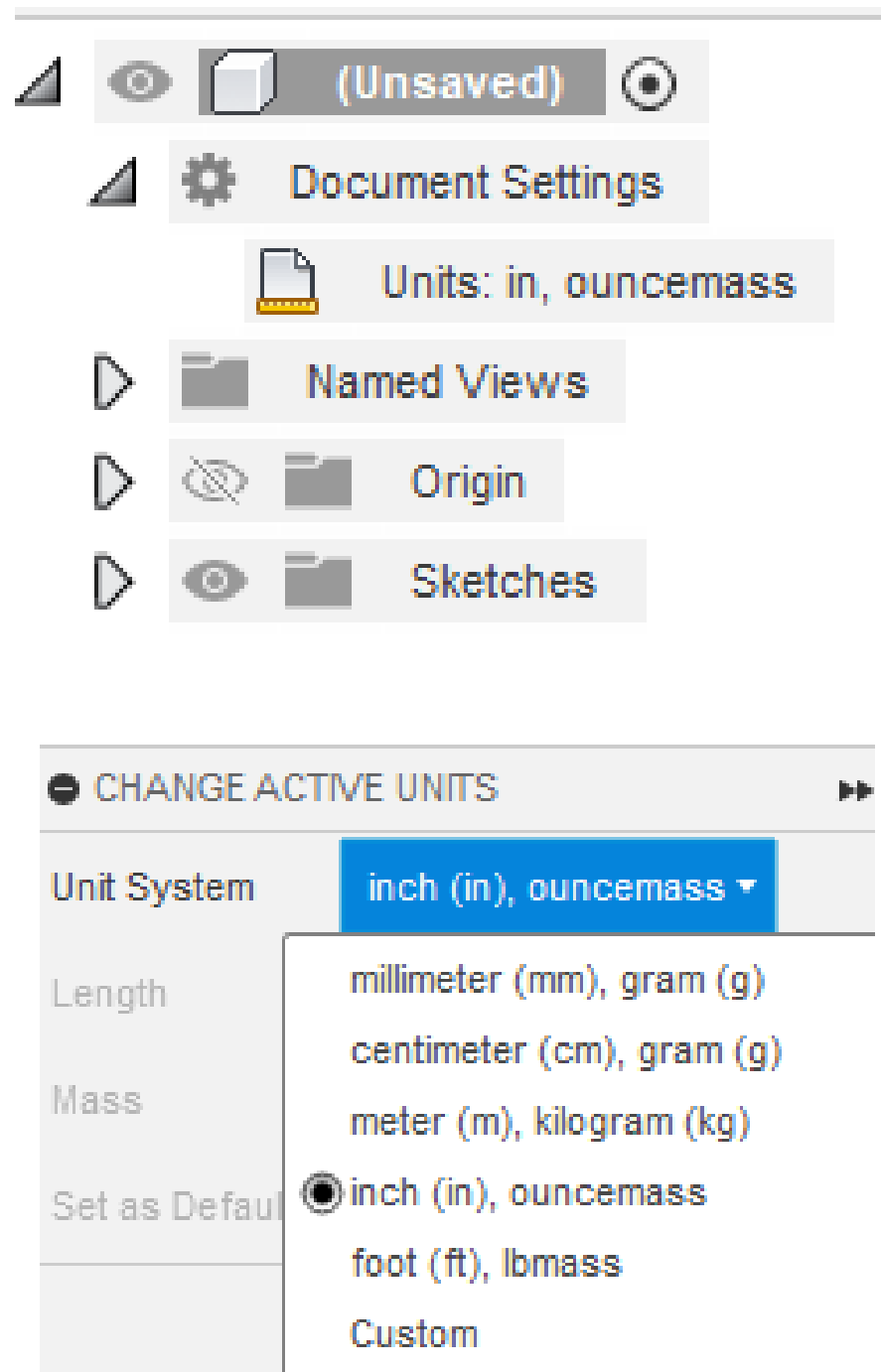
Fusion is located on a 3d plane, so you'll be able to view 360° around your model. The cube in the top right corner shows you what planes you are viewing, and the home button will take you back to the base view. Dragging with the scroll wheel will allow you to pan around your model, while holding shift + scroll wheel will allow you to orbit around your model. If you're not using a mouse, you can switch between the views with the icons on the bottom screen; however, it is highly recommended to use a mouse.





Dimensions

When modeling components for real-life, it's important to use exact measurements to ensure each component functions as intended with one another. Fusion grants us the ability to input meticulous measurements to ensure that our designs can have near-perfect accuracy. In the top left corner of your screen, you can open up a window under document settings to change your project's dimensions to your preference.



Project 1: Base Plate

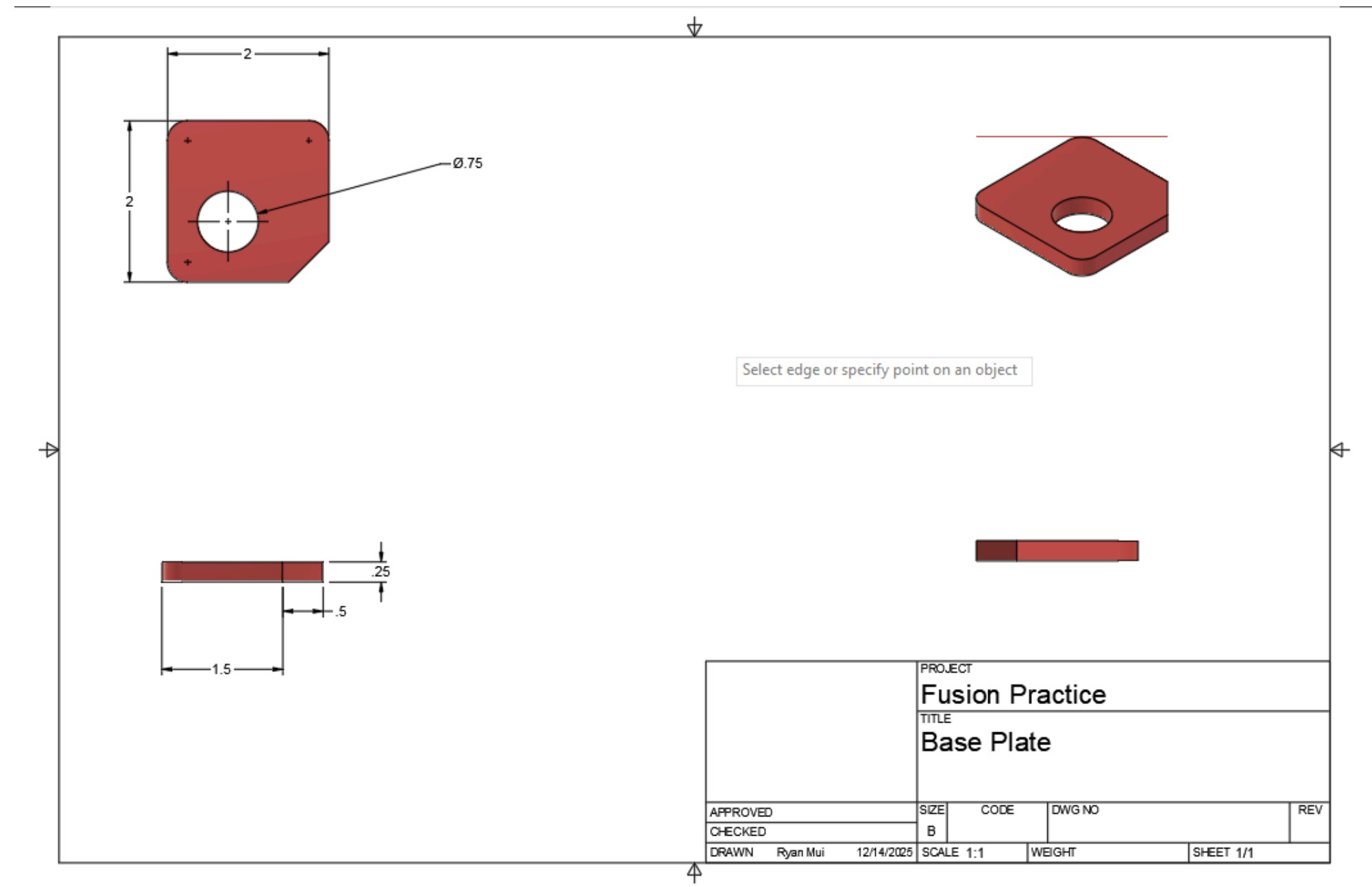
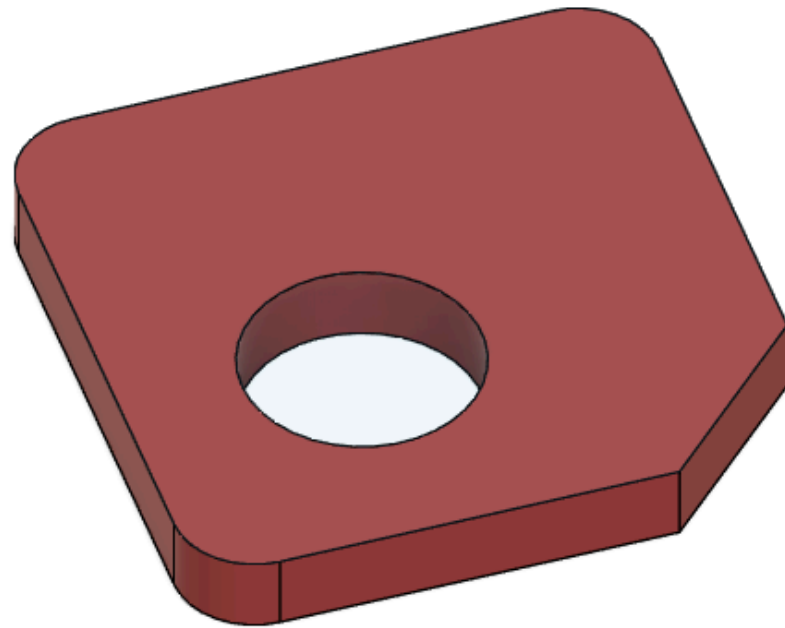
This Project Teaches:

1. Sketching on the XY plane
2. 2-point rectangles
3. Dimensioning with constraints
4. Fillets vs chamfers
5. Extruding sketches into solids
6. Applying appearance





Base Plate

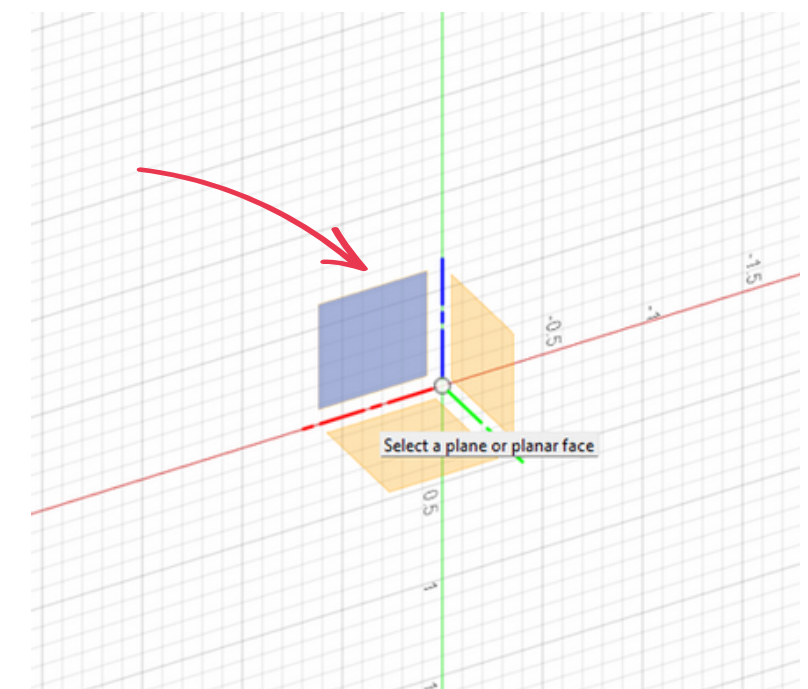
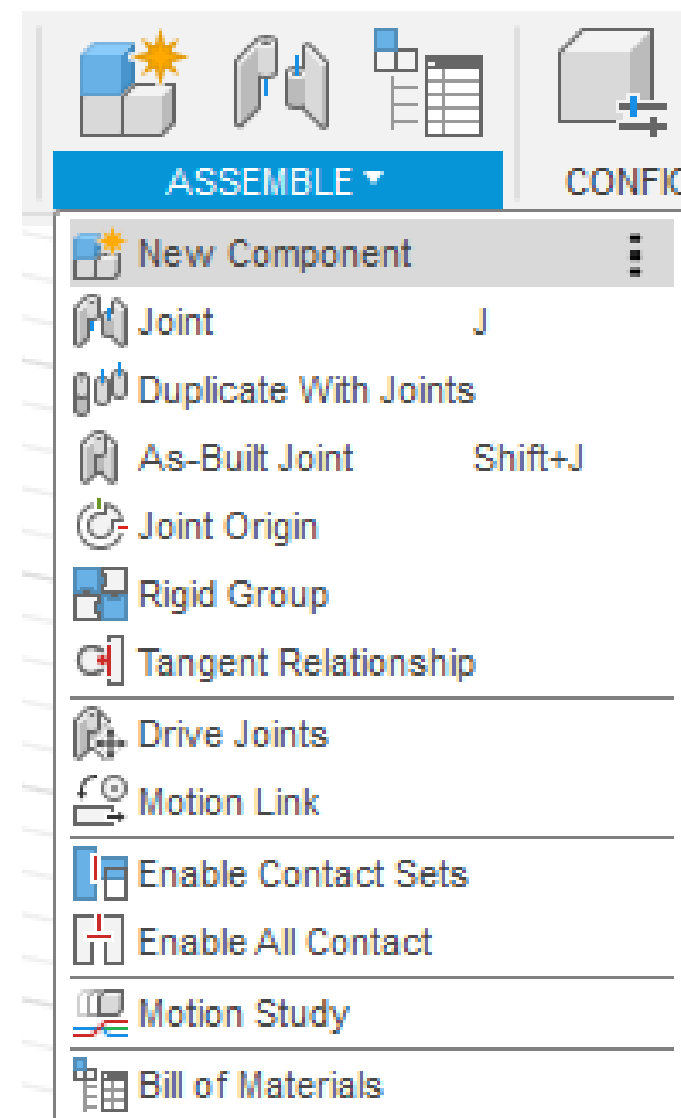




Starting a Sketch

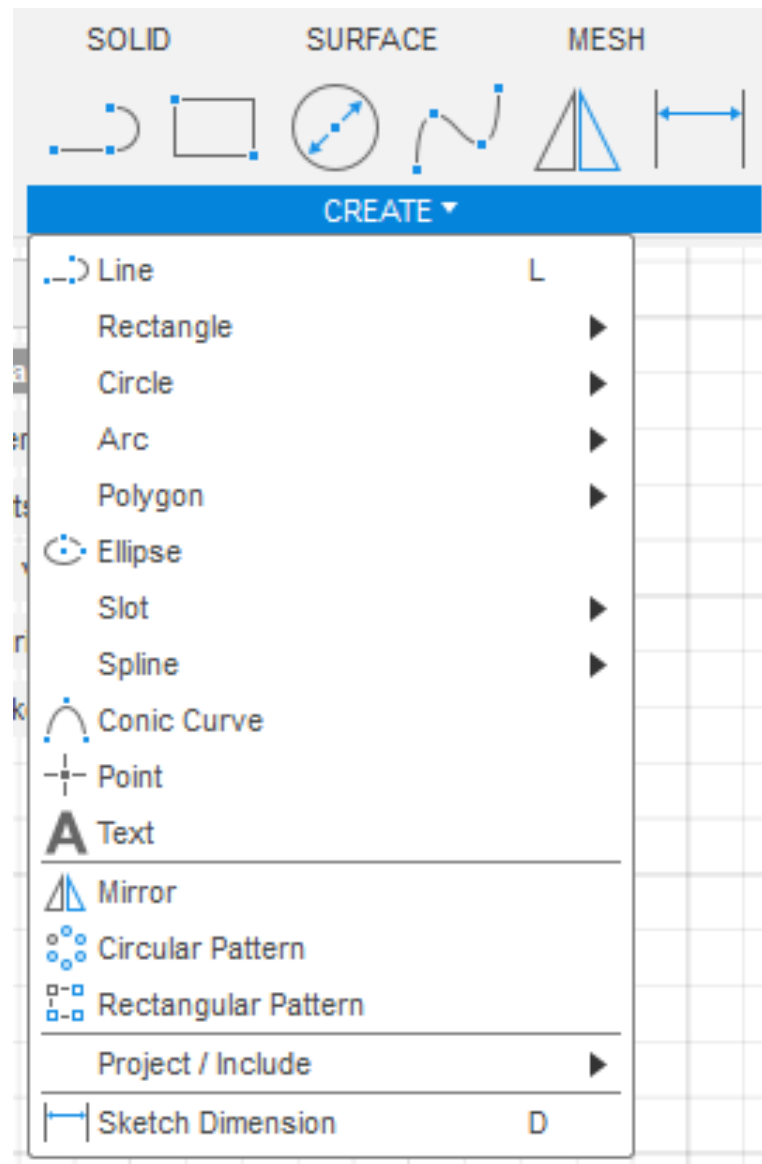
The first thing we want to do is make a new component, which is located under the assemble tab.

We are going to start by selecting the **XY plane**, the plane pointed at in the photo on the right. This should focus your camera on the plane as we build the base of our model.



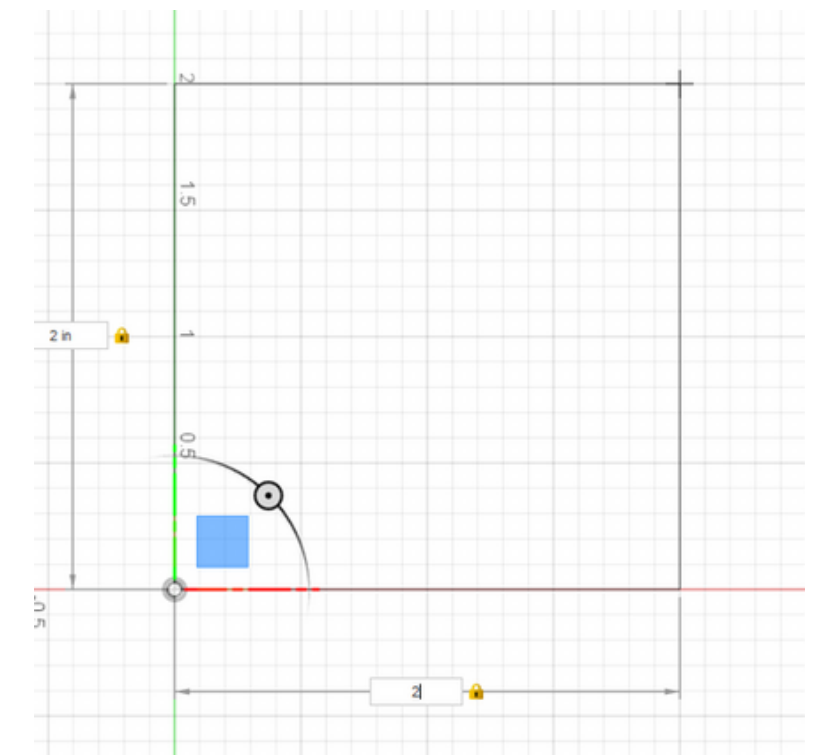


Making a Rectangle



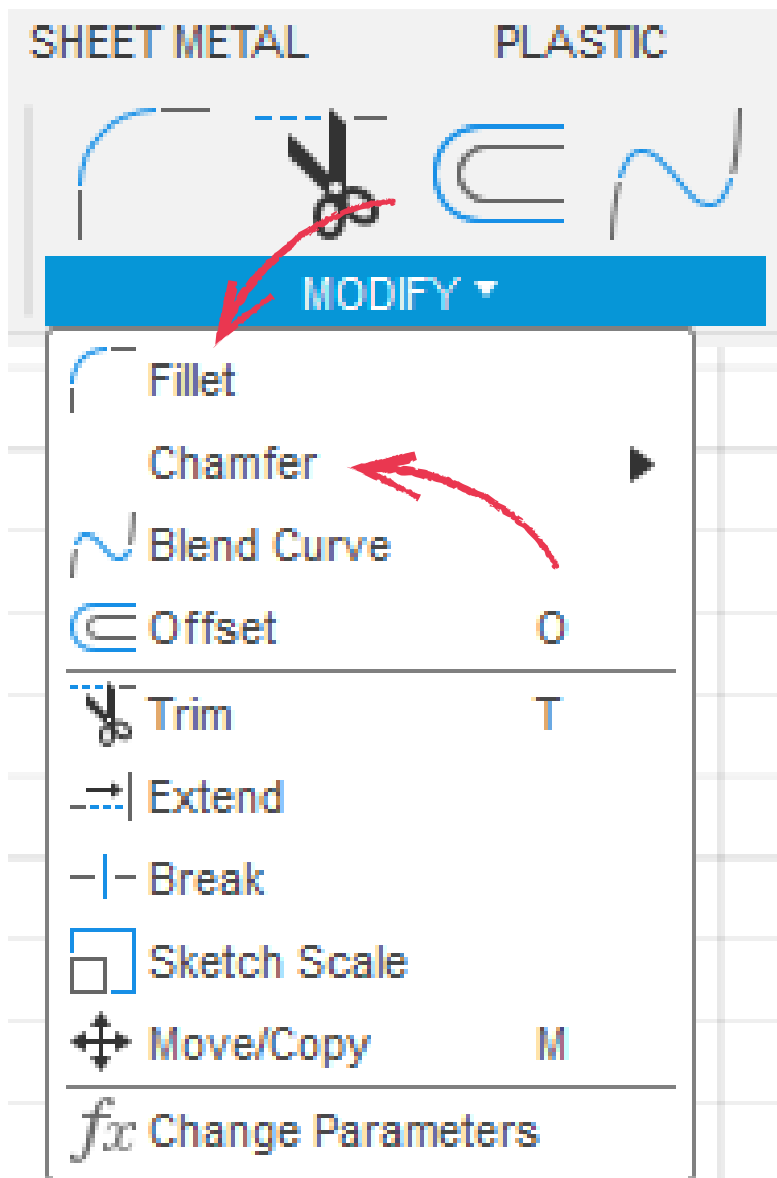
When in sketch mode, Fusion will open up a new list of sketch-specific tools for us to use. We can start by selecting a rectangle by selecting the tool in the drop-down menu, or by clicking the shortcut “r”. Use the basic two-point model, and start your first click at the origin. Notice the boxes on the length and the width? Those indicate the dimensions. We are going to make a 2x2 square for the base.

You can hit tab to switch between the length and width dimensions.

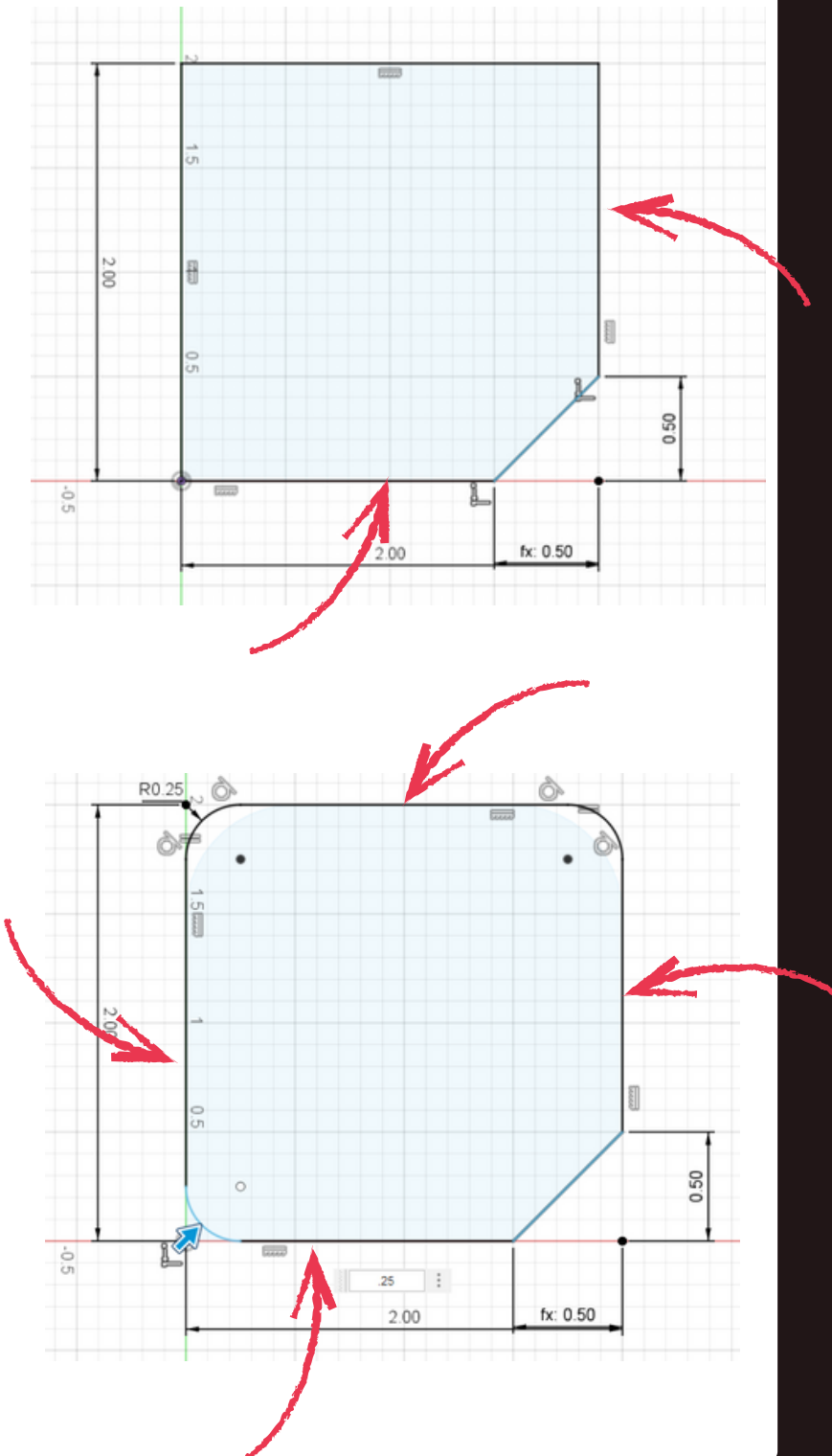




Fillet & Chamfer 3d



We are now going to learn how to fillet and chamfer a sketch. Start by opening the modify drop-down to access our tools. Let's start by creating a chamfer on the bottom right corner of our base. Select the chamfer tool, select equal distance chamfer, select the bottom and right edges, and enter a chamfer of .5 in. To select multiple sides at once, you can hold down Shift and select them. We will now do something similar with the fillet tool. Select the fillet tool, select all the sides of the base, and enter a fillet of .25 in.



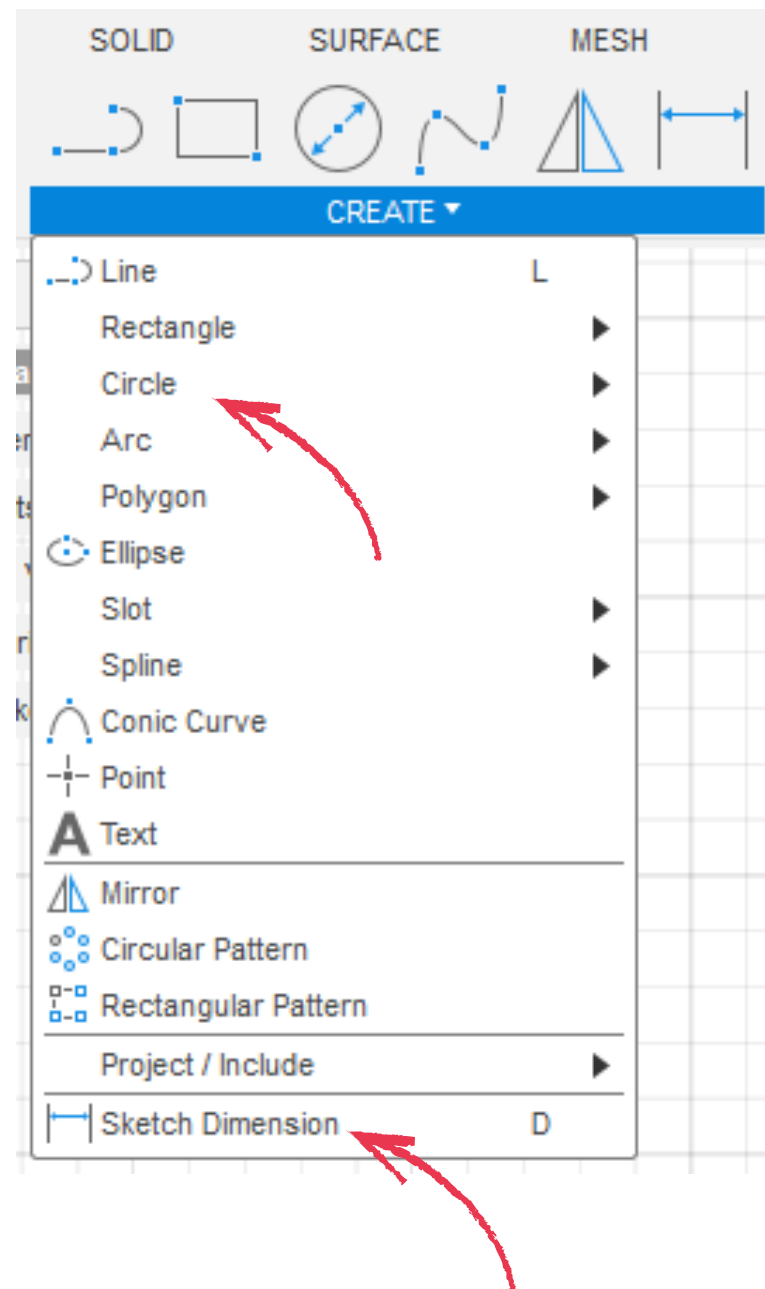


Important Note

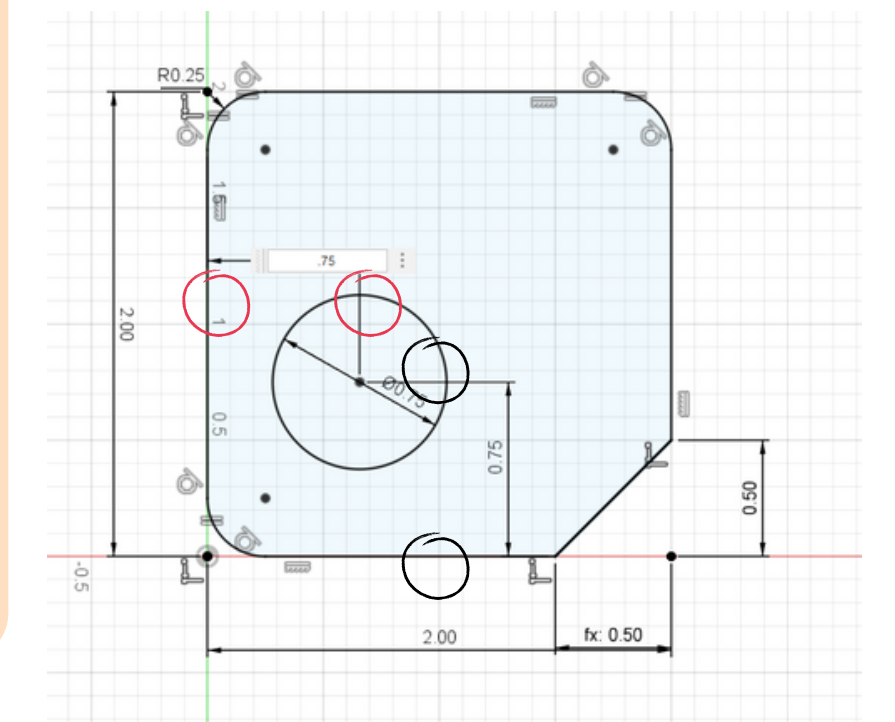
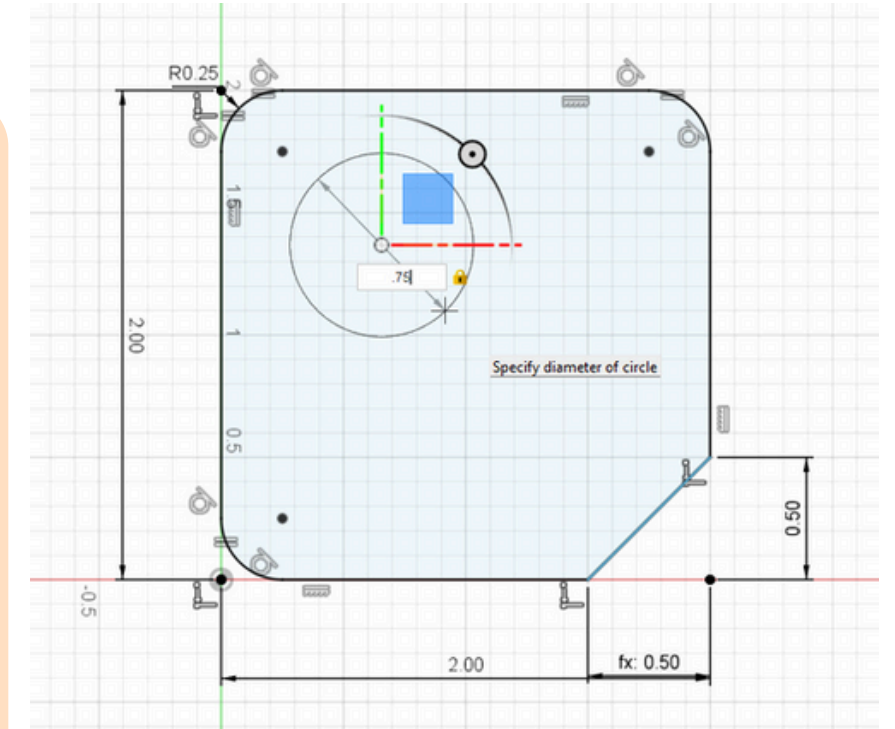
Make sure to not hit finish sketch. We want to do all these actions under one sketch. You can also fix/redo a mistake by hitting ctrl z.



Dimension Tool

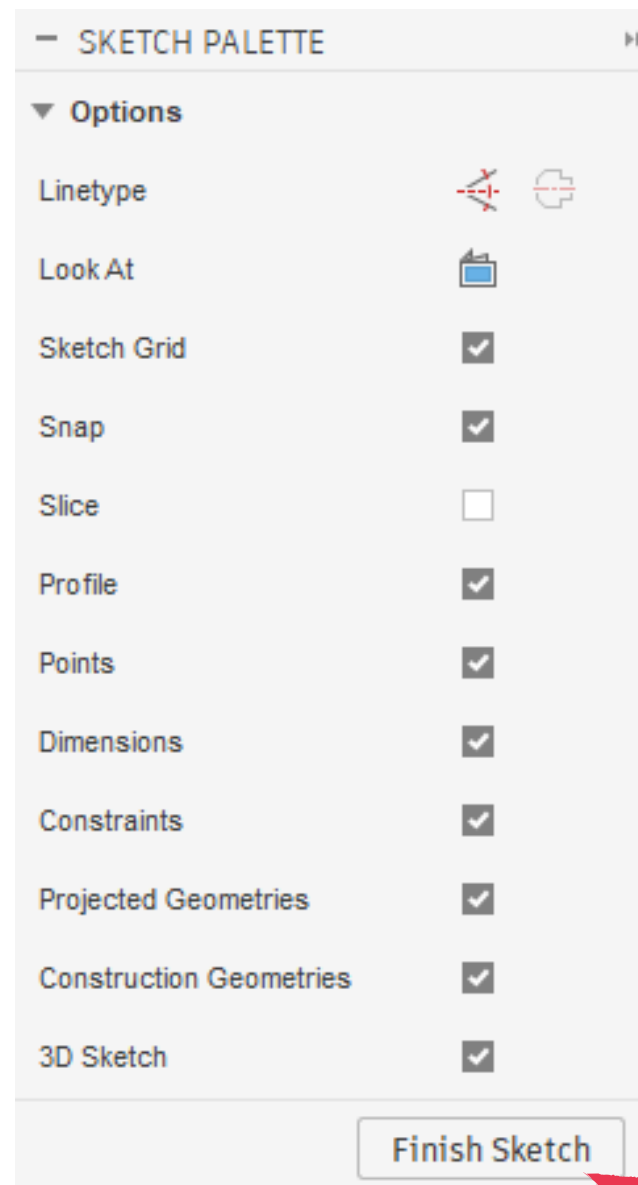


Now we are going to use the circle tool. Place the circle anywhere on the base, and enter a radius of .75 in. We can place the circle anywhere, as we'll arrange its specific location with the dimension tool. Select the dimension tool, or use shortcut d. Look at the second model below. The colored circles show how we are going to reposition the circle on the x-axis and the y-axis. Start by selecting the edge of the base and the specified point on the circle. Click to lock the dimension sketch, and enter .75 in. Do the same for the other side.

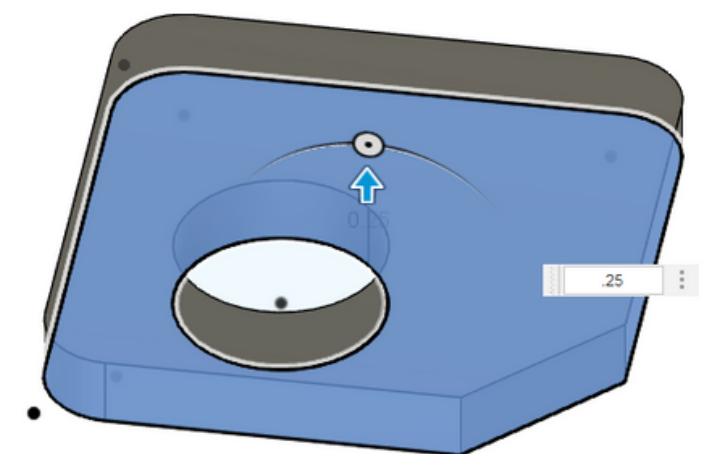
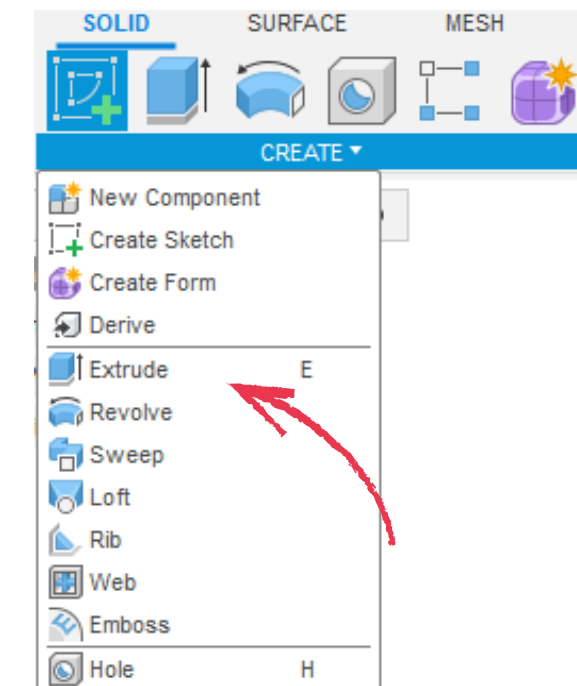




Extruding Sketches

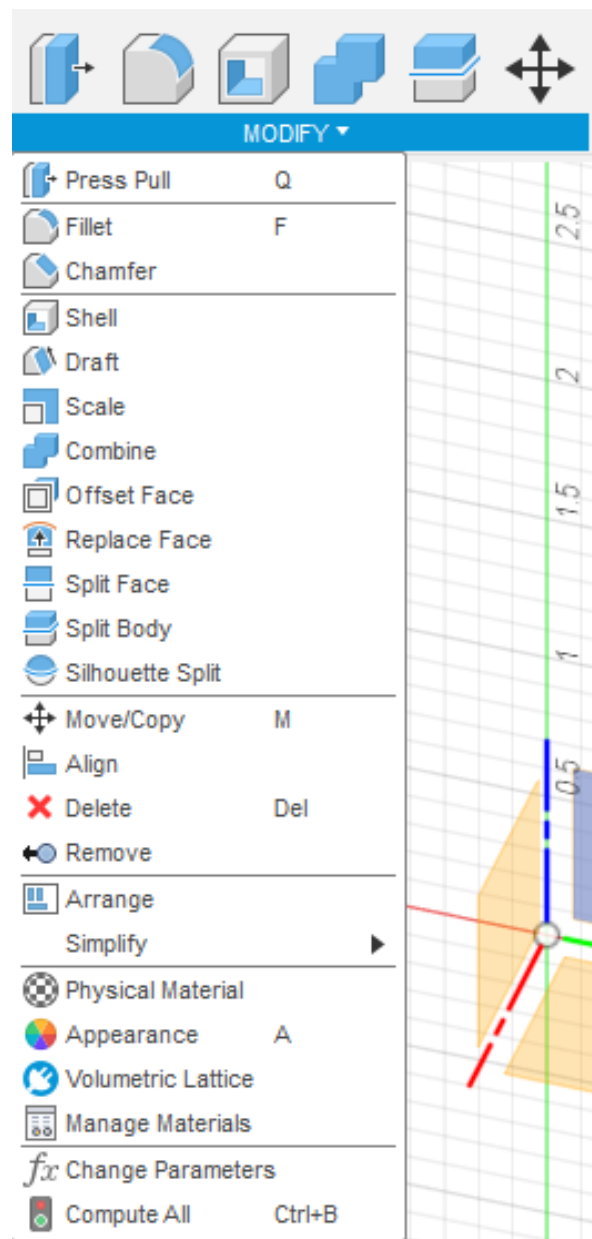


We can now hit finish sketch!
Next, open the drop-down menu under create and select the extrude tool, or shortcut e. Select the face of our base and enter .25 in. Make sure the arrow is pointing upward so that our extrusion is upward. We can hold the arrow and drag up or down to change its direction.

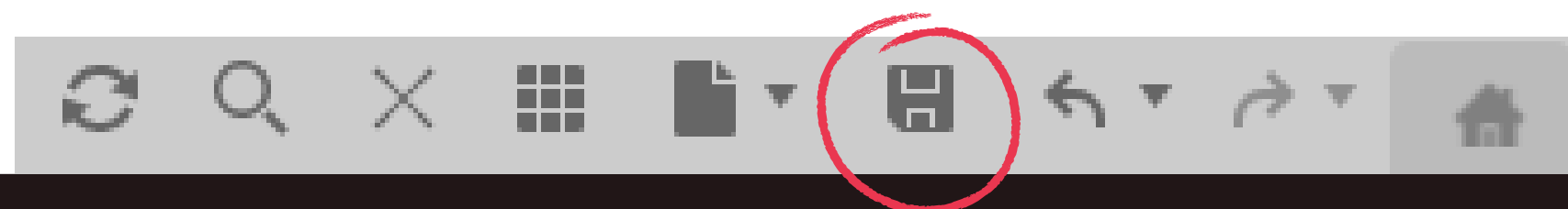
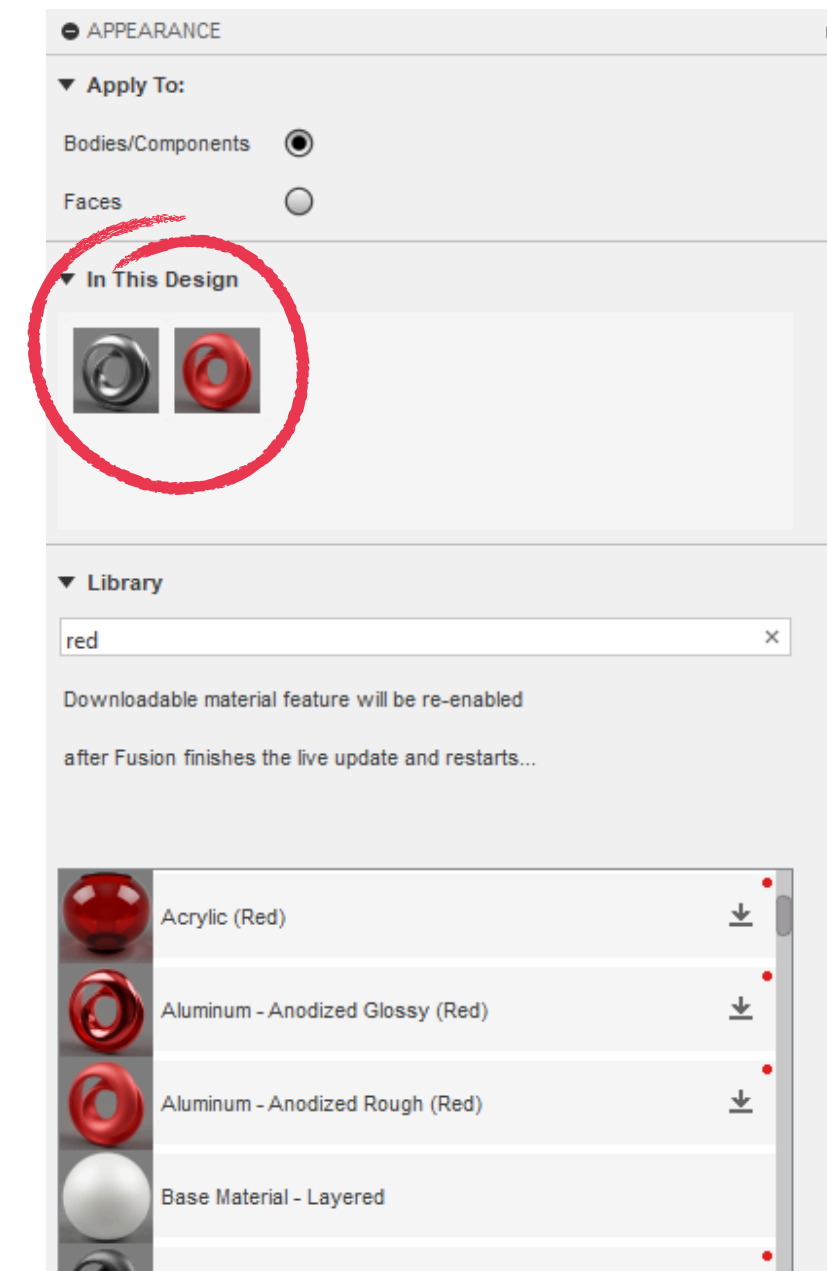




Colors/Appearance



Lastly, let's add some color to our model. Under the modify drop-down, we can select appearance or shortcut a. Search for whatever color you'd like and hold and drag it into the top box as shown. To add the color, make sure your desired color comes first in the order. We can now save our model by selecting the tool on the top bar of your screen.



Project 2: Bracket

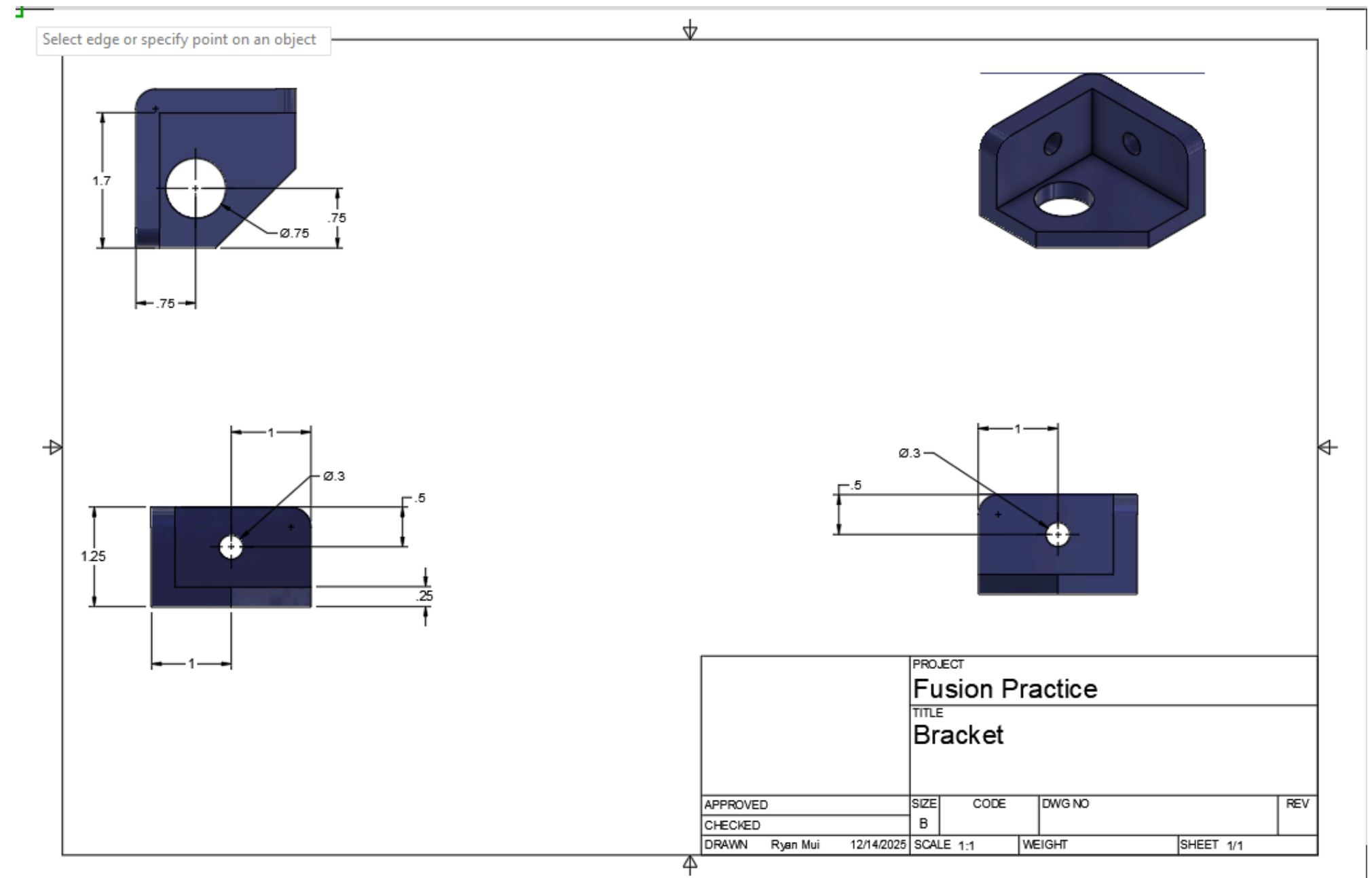
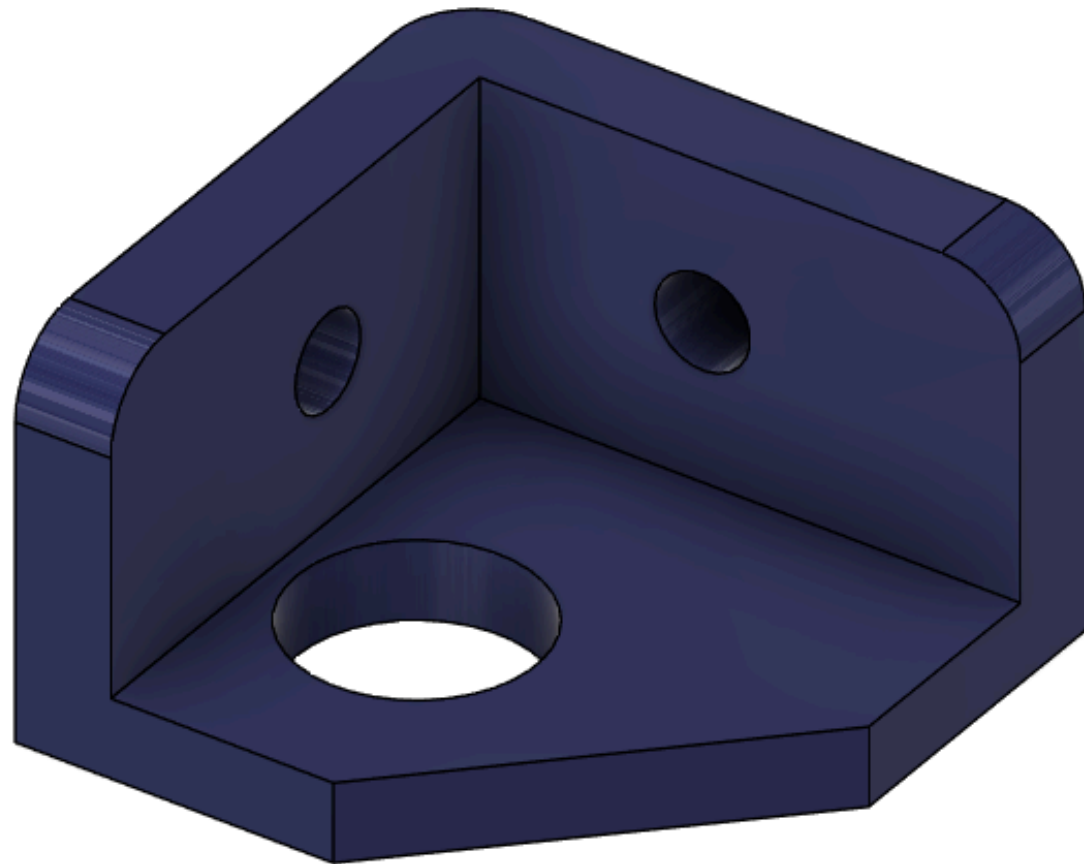
This Project Teaches:

1. Making sketches on existing faces
2. Extruding multiple profiles
3. 3D fillets and chamfers
4. Creating functional holes
5. Designing for attachment/support



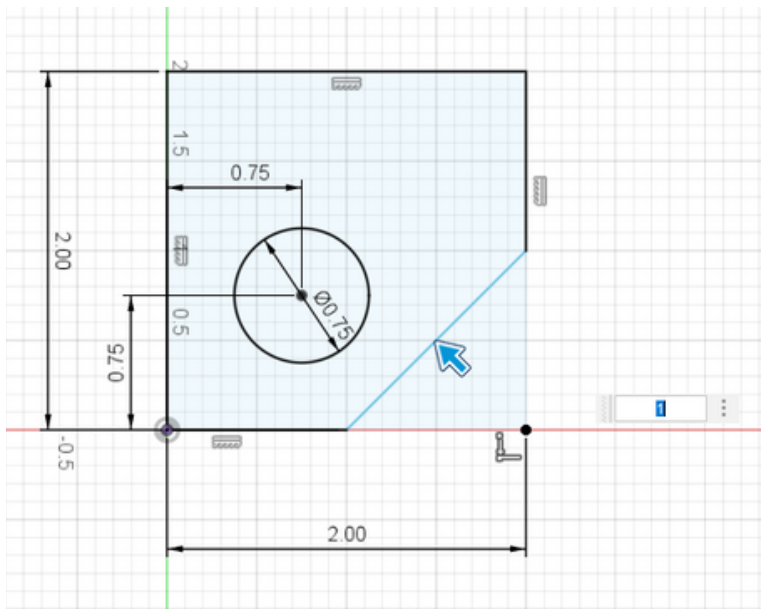


Bracket

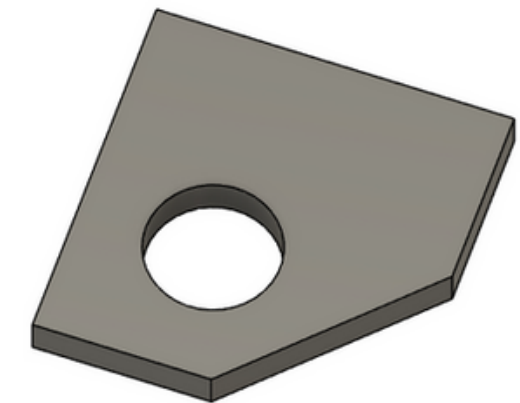




Making New Design



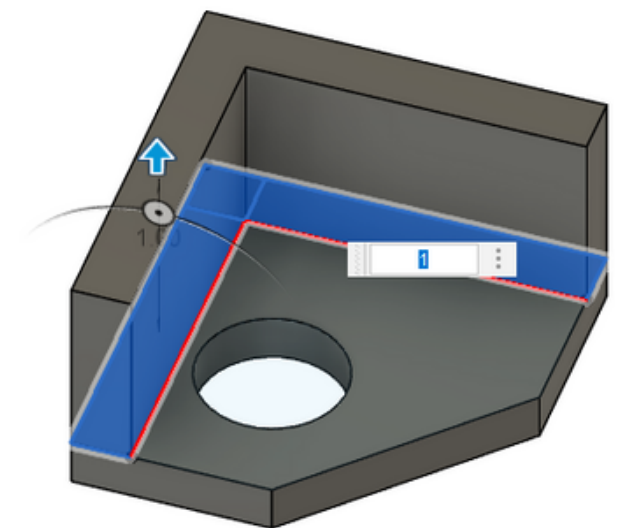
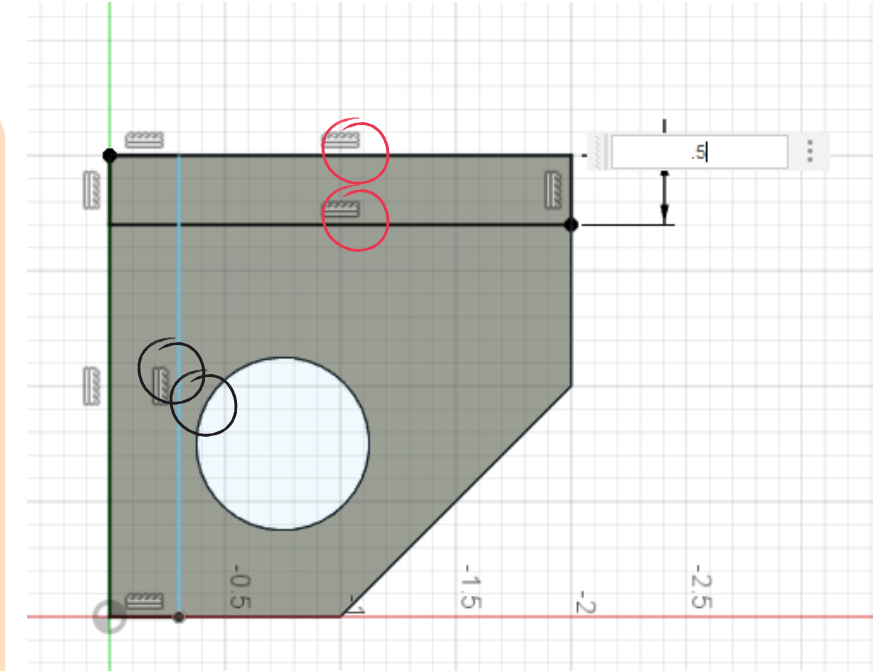
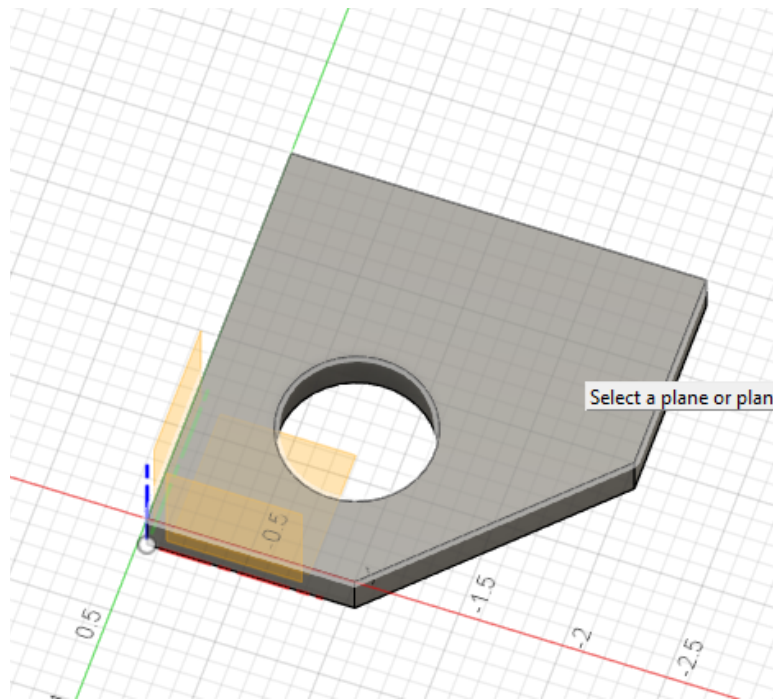
We're going to make a new design, so click the plus at the top bar on your screen. Start again by making a new component. The base of the bracket will be quite similar to the base plate's, so repeat the same steps to make your base sketch; however, don't fillet the sides and make the chamfer 1 in instead of 0.5 in. The hole will remain in the same position. Once again, finish the sketch, extrude it .25 in.





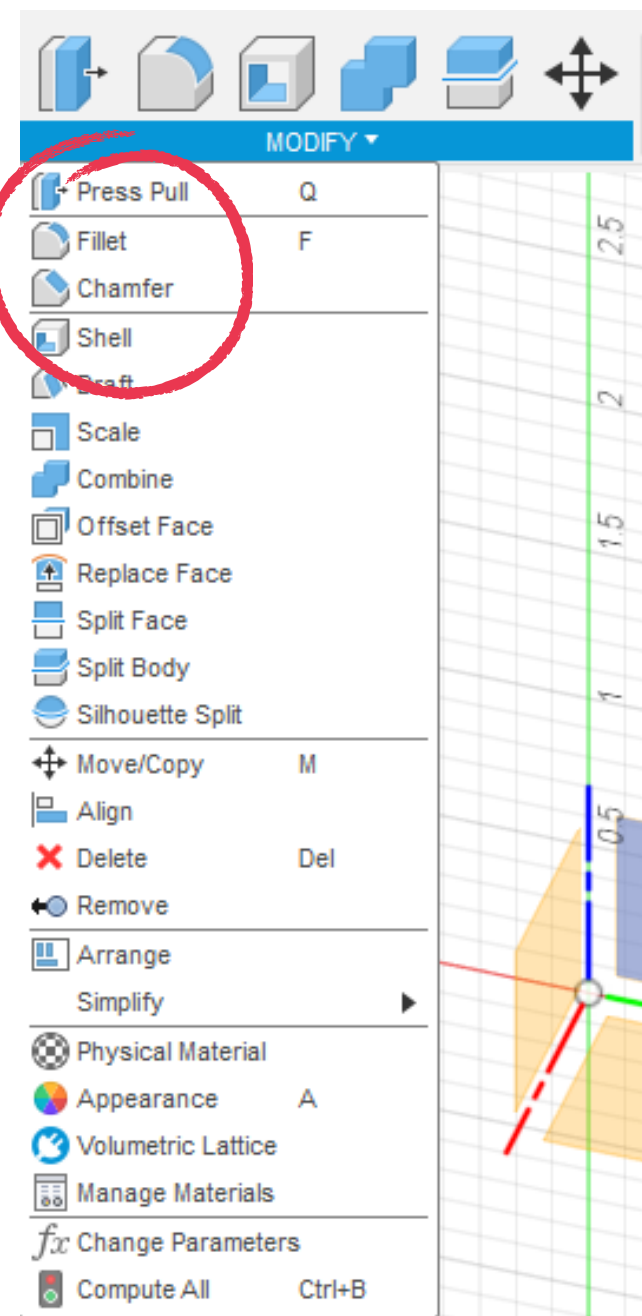
Sketching on Faces

Fusion allows us to make sketches on our existing models, so we select the top face of our bracket's base. Make two rectangles as shown in the picture on the right (dimensions don't matter for now) and then use the dimension tool to make them .5 in from the edges. Finish your sketch, and then extrude the two rectangles 1 in up. Make sure to select all of the sketch, including the overlapping section in the top left corner.

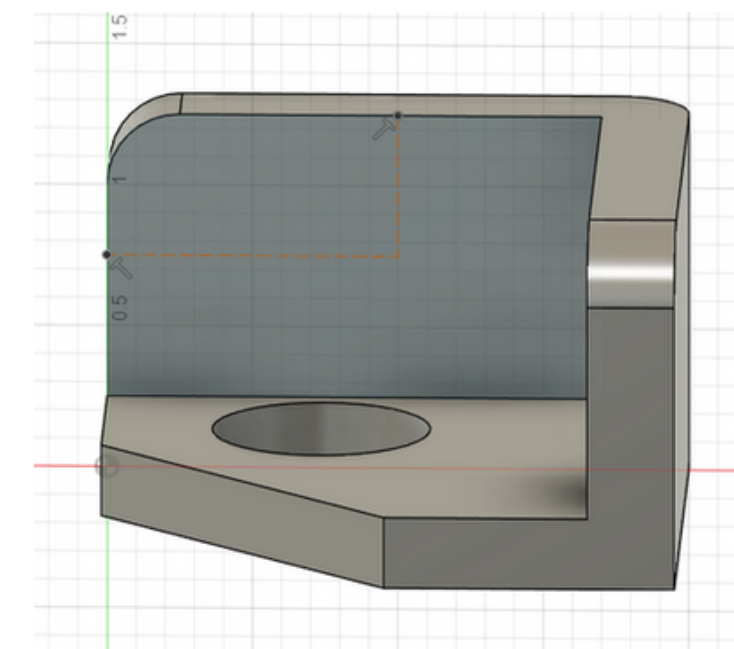
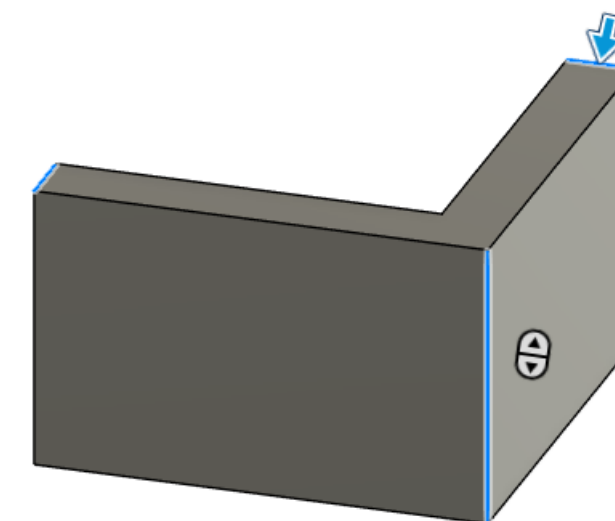




Fillet & Chamfer 3d



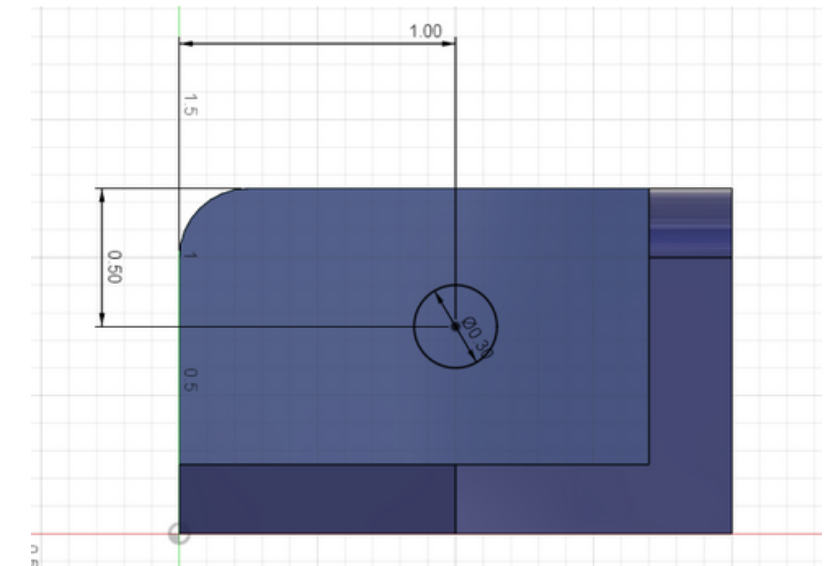
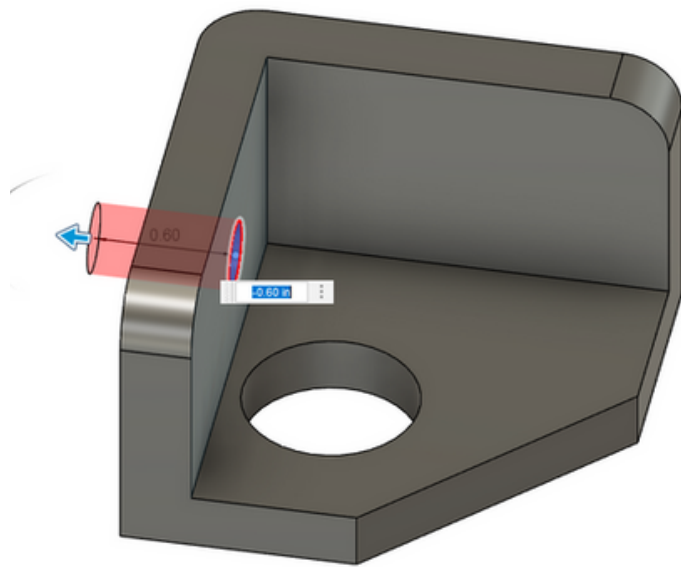
As you might've seen, the fillet and chamfer tools also exist outside of sketches. The fillet/chamfer tools allow you to make cuts and curves on 2d sketches, while the versions under the modify drop-down allow you to do the same with your models. Select the sides shown in the picture on the right, and enter a fillet of .25 in. We won't be using the 3d chamfer tool, but know it does the same to models as it would with your sketches.





Making Holes

Now we are going to make holes to attach our later components. Create a new sketch on one of the inner vertical faces of the bracket and make a circle with a diameter of 3 in. Use the dimension tool to position the circle .5 in from the top and 1 in from the outward side. Next, extrude the circle; however, pull the arrow backward to make a cut-out. Drag until the circle fully cuts through. You can repeat these steps on the other inner face. Lastly, apply a color of your choice to the bracket. Don't forget to save your sketch.

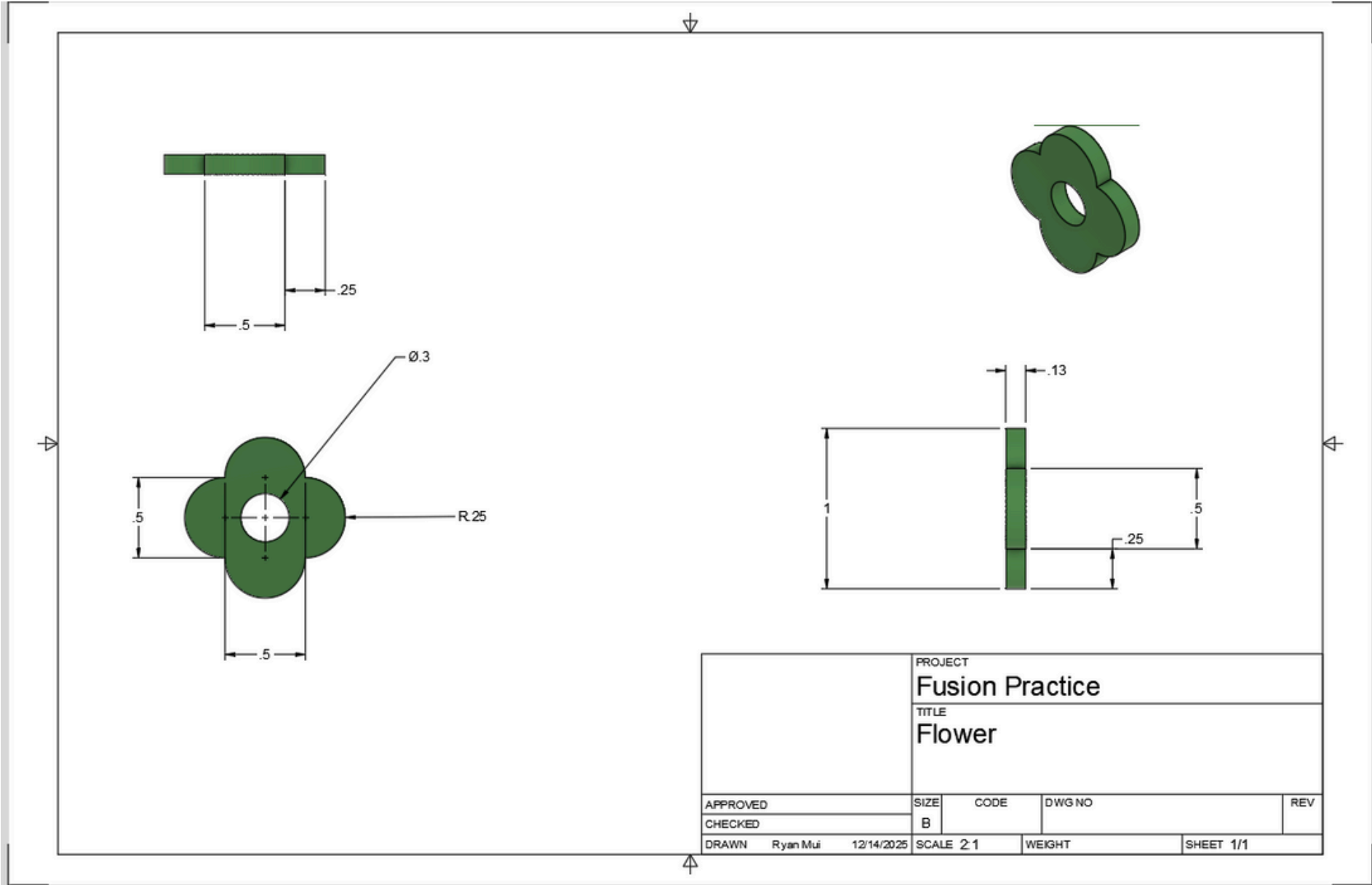
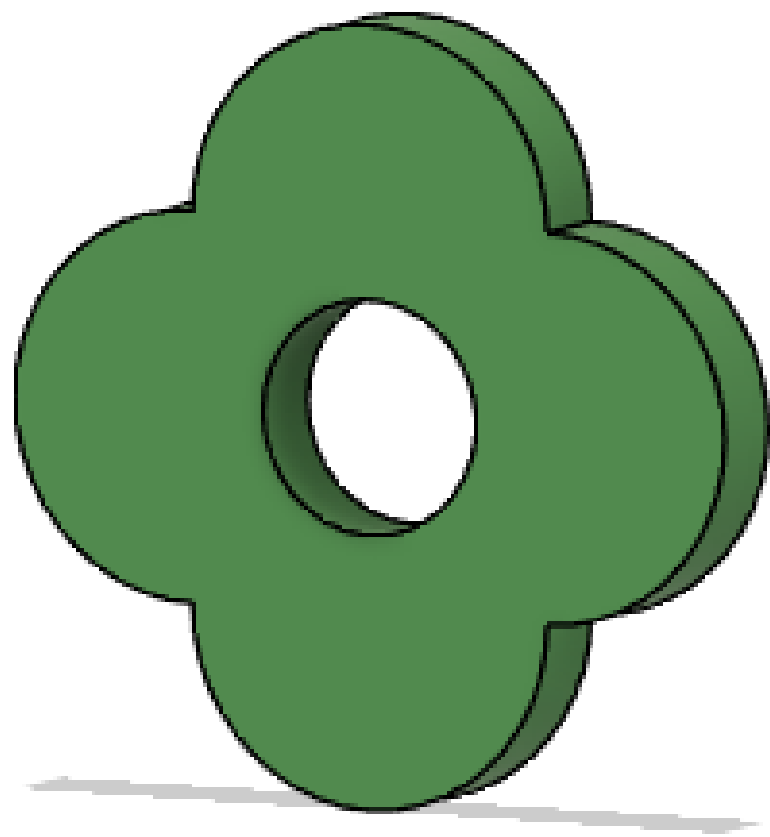


Project 3: Flower

This Project Teaches:

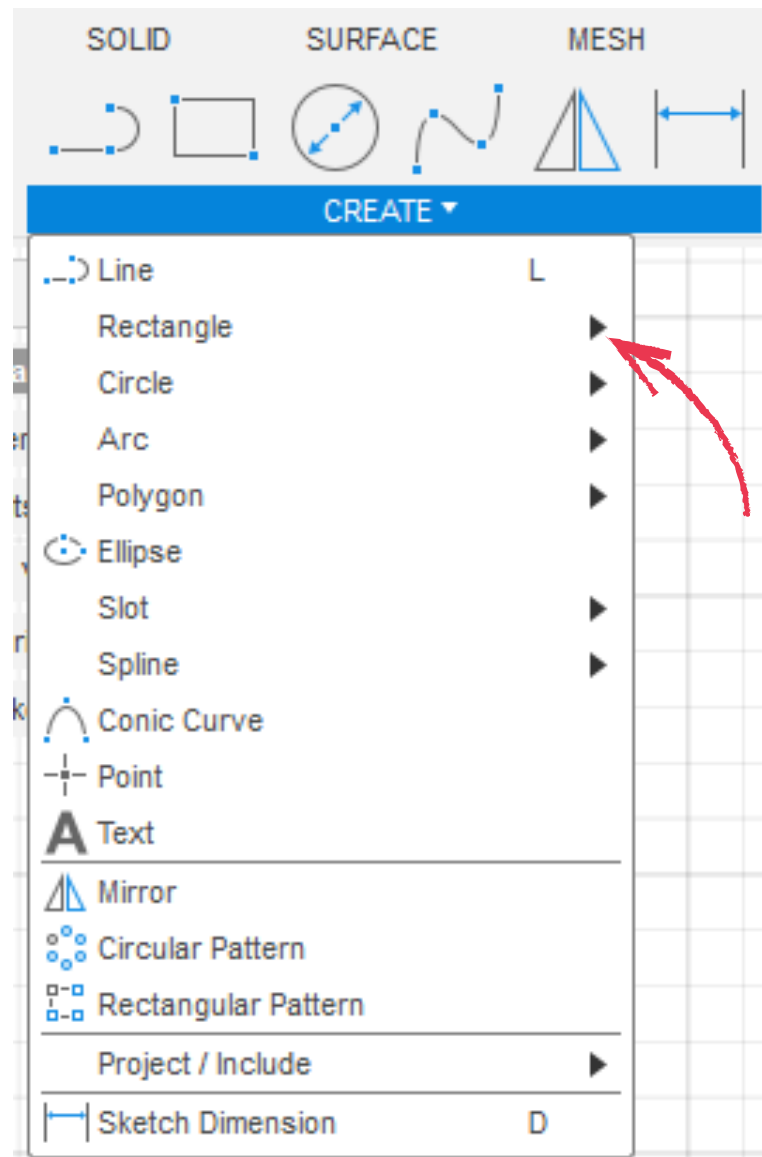
1. Center rectangles
2. Construction lines as scaffolding
3. 3-point arcs
4. Trimming sketches
5. Symmetry and radial design thinking



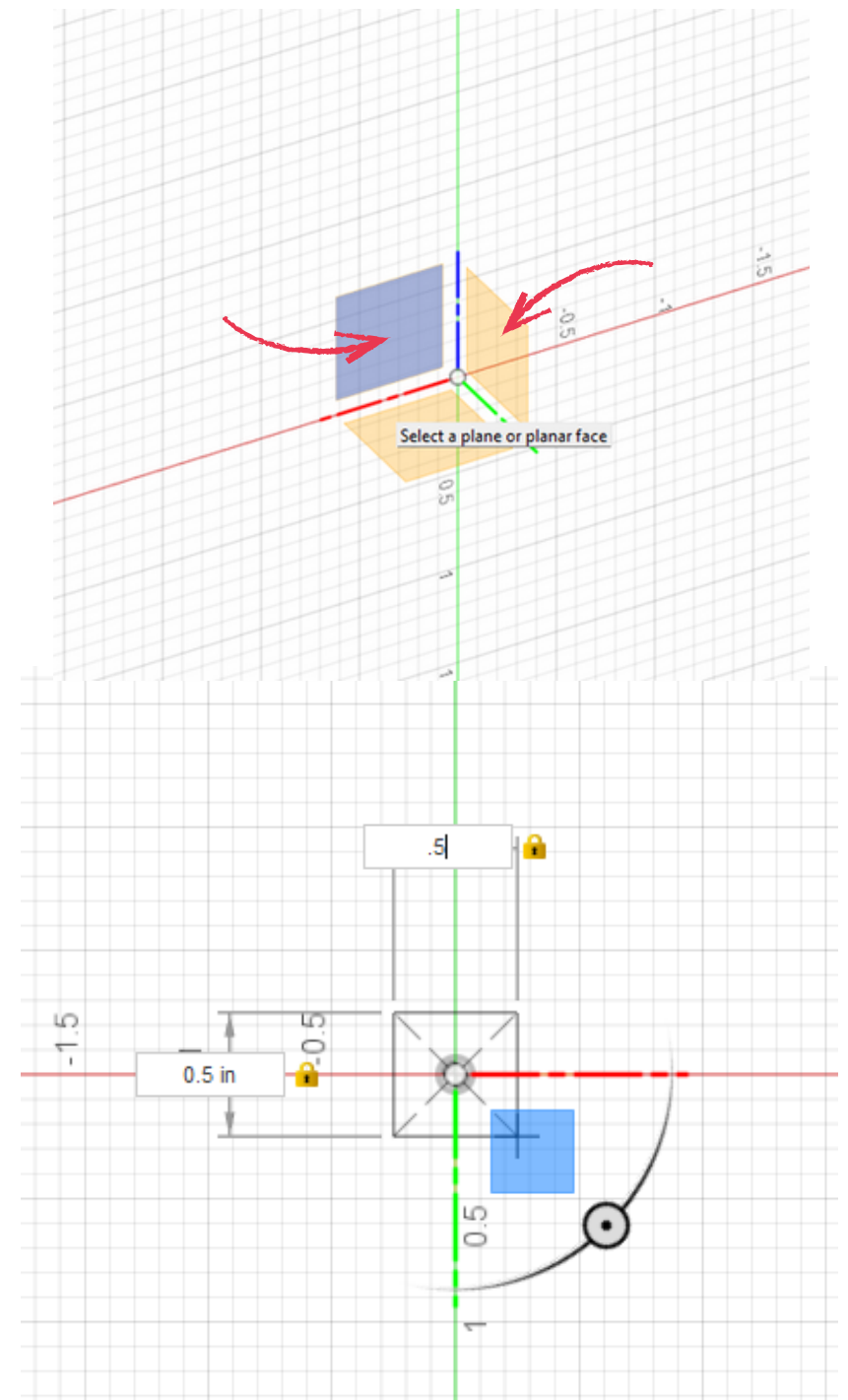




Different Rectangles

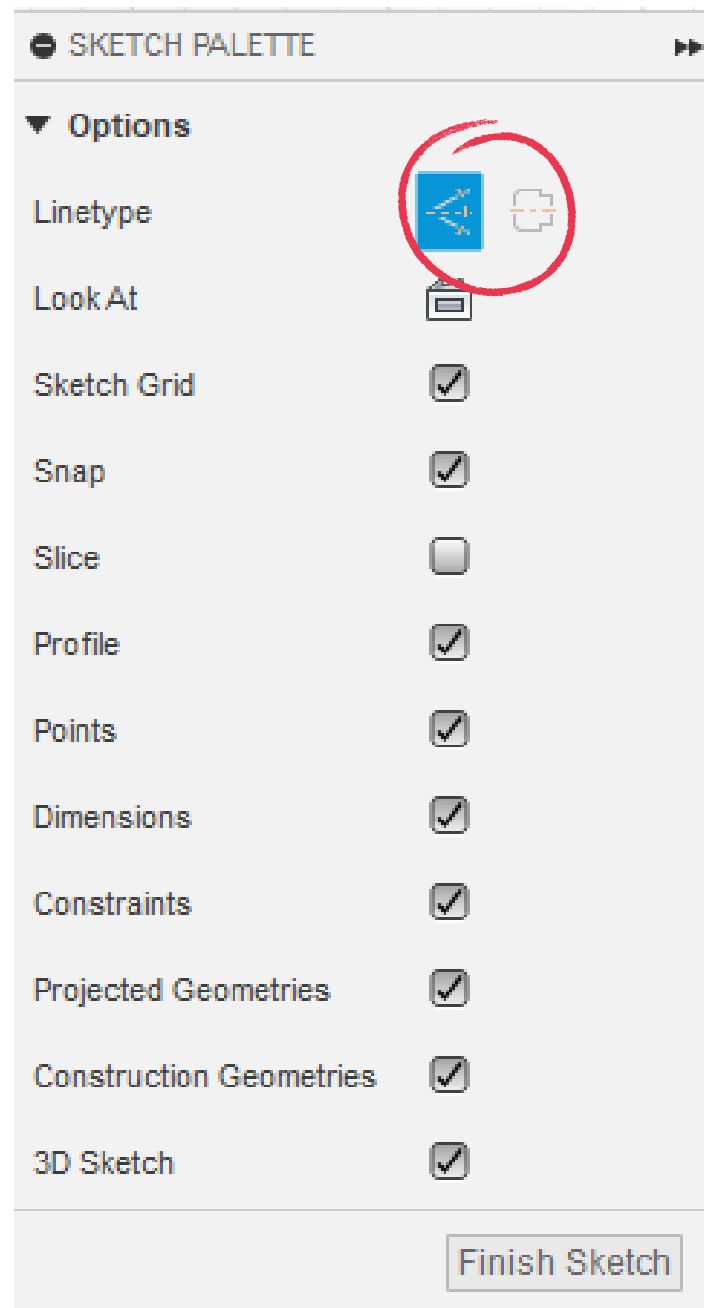


We are going to make another new model, so open up a new sketch and make a new component. This time we'll be making a vertical sketch, so select either the XZ or YZ plane. Instead of making a 2-point rectangle, we are going to make a center rectangle. Click the arrow next to the rectangle tool and select "center rectangle" in the pop up. Click the origin of your grid and set the length and width to 0.5 in.



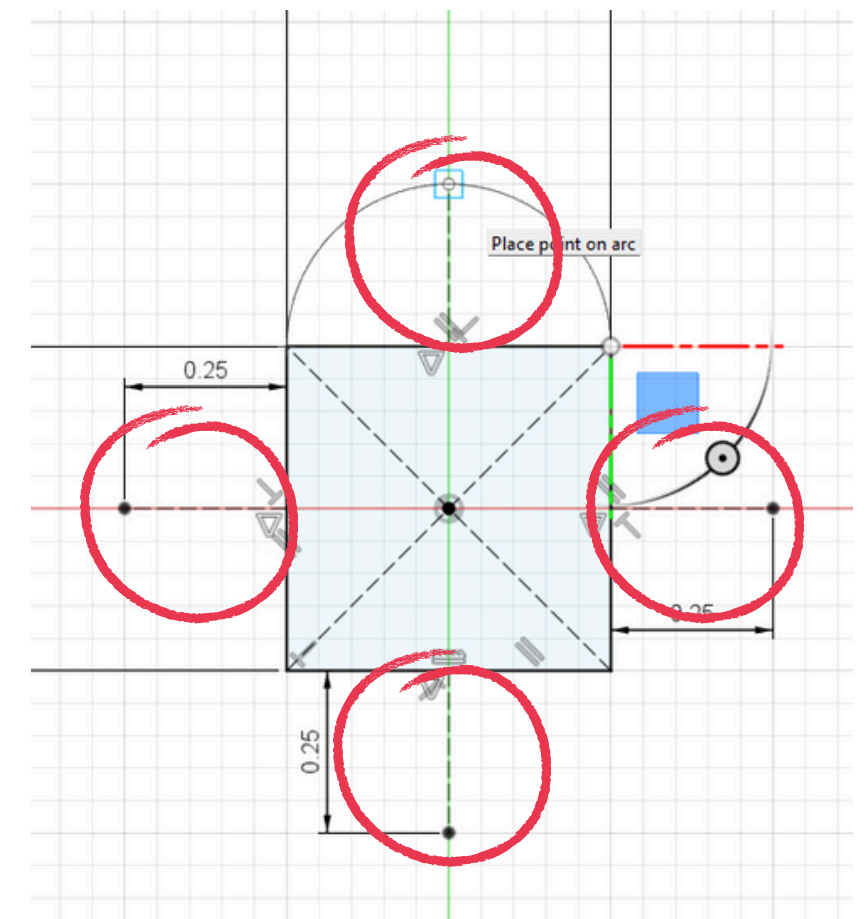


Construction Lines



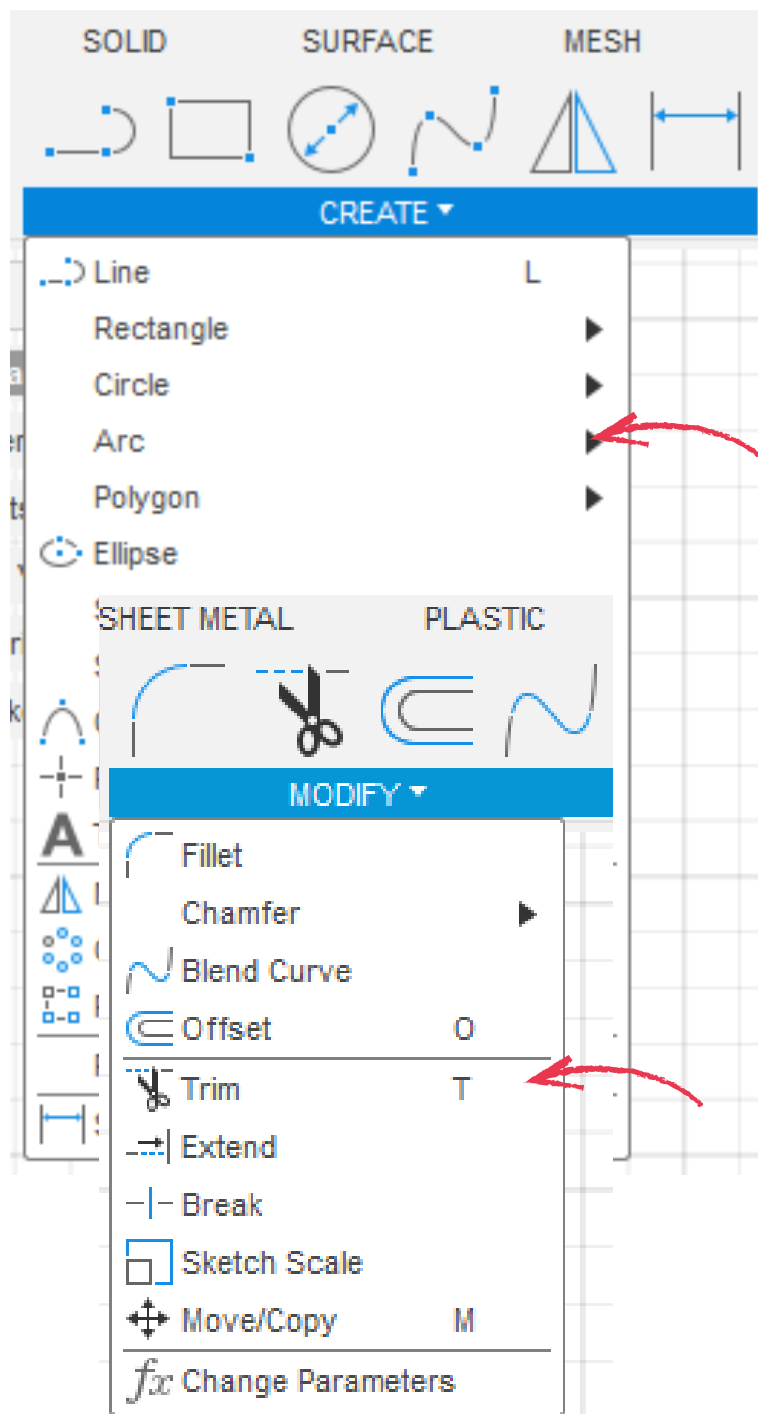
Construction lines are placeholder lines we can make to serve as the scaffolding to our models. They make outlines/lines that we can snap to and easily erase when we're done. Select the line tool, or shortcut L, and switch the linetype to construction in the right-side menu. From the mid points of each side of the square, make a line 0.25 in long from each face.

Note: Setting your linetype to construction lines will make ALL sketches construction lines. Make sure to deselect construction when you are done



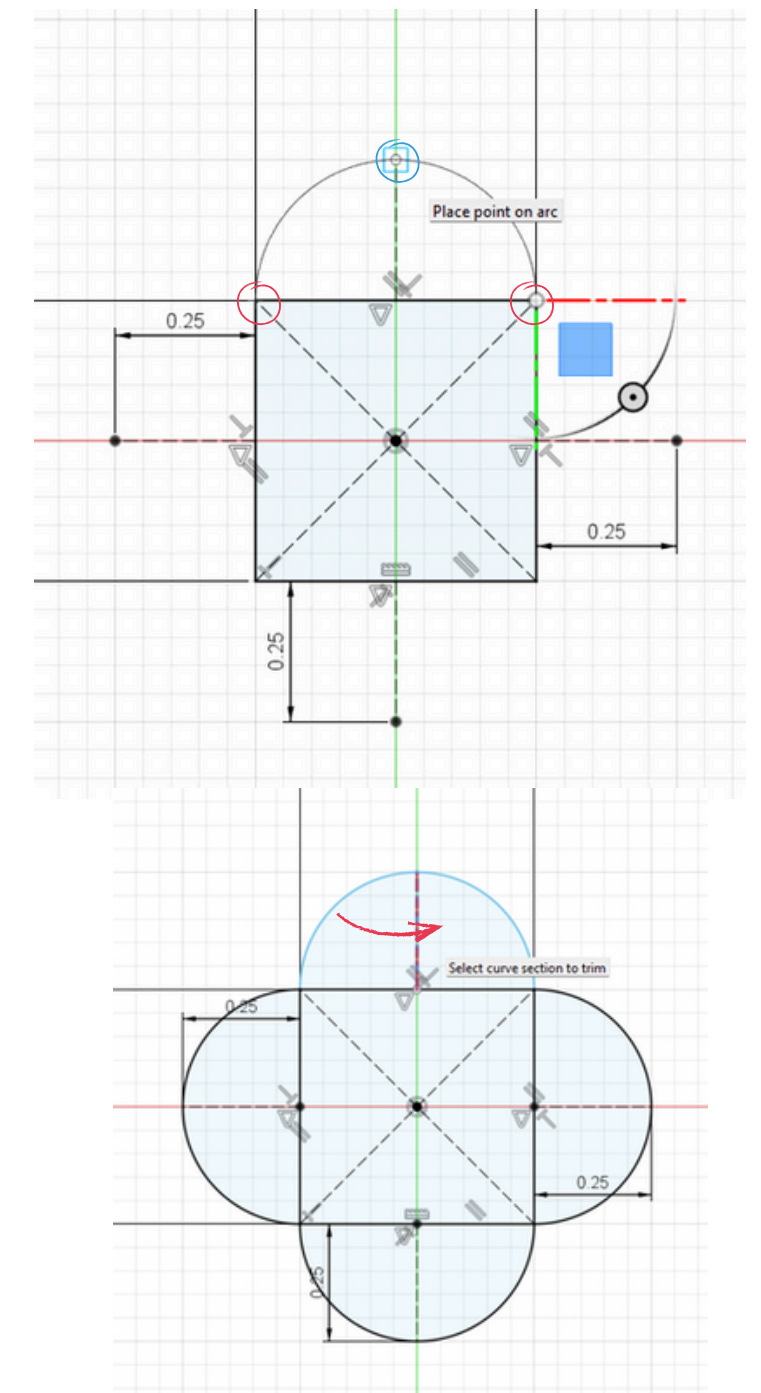


Arcs and Trim Tool



Now let's make the petals of our flower. Under our create drop-down menu, select the arc tool and select 3-point arc. Follow the picture on the top right.

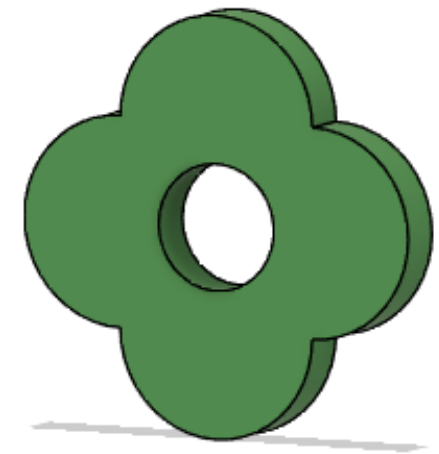
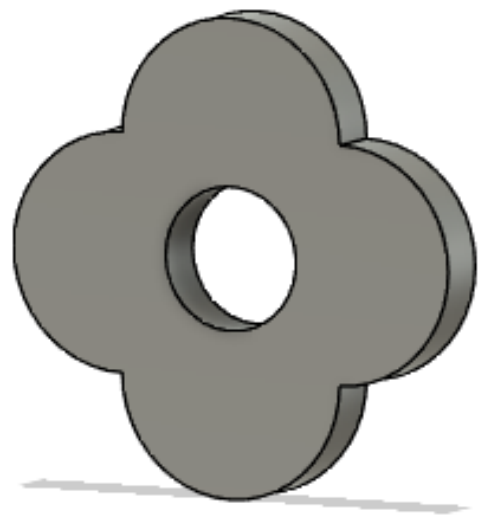
First, click the two corners of the circle, and snap the midpoint of your arc to the endpoint of your construction line. Then, repeat the steps for the other 3 sides of the square. Lastly, open the trim tool under the modify drop-down menu or shortcut T. Highlight each of the construction lines and enter to remove them from your sketch.





Finishing the Flower

Let's quickly finish up the flower. Create a circle at the origin with a diameter of 0.3 in and finish our sketch. Extrude the sketch .125 in and add colors. Make sure to save once you are done.



Project 4: Bolt

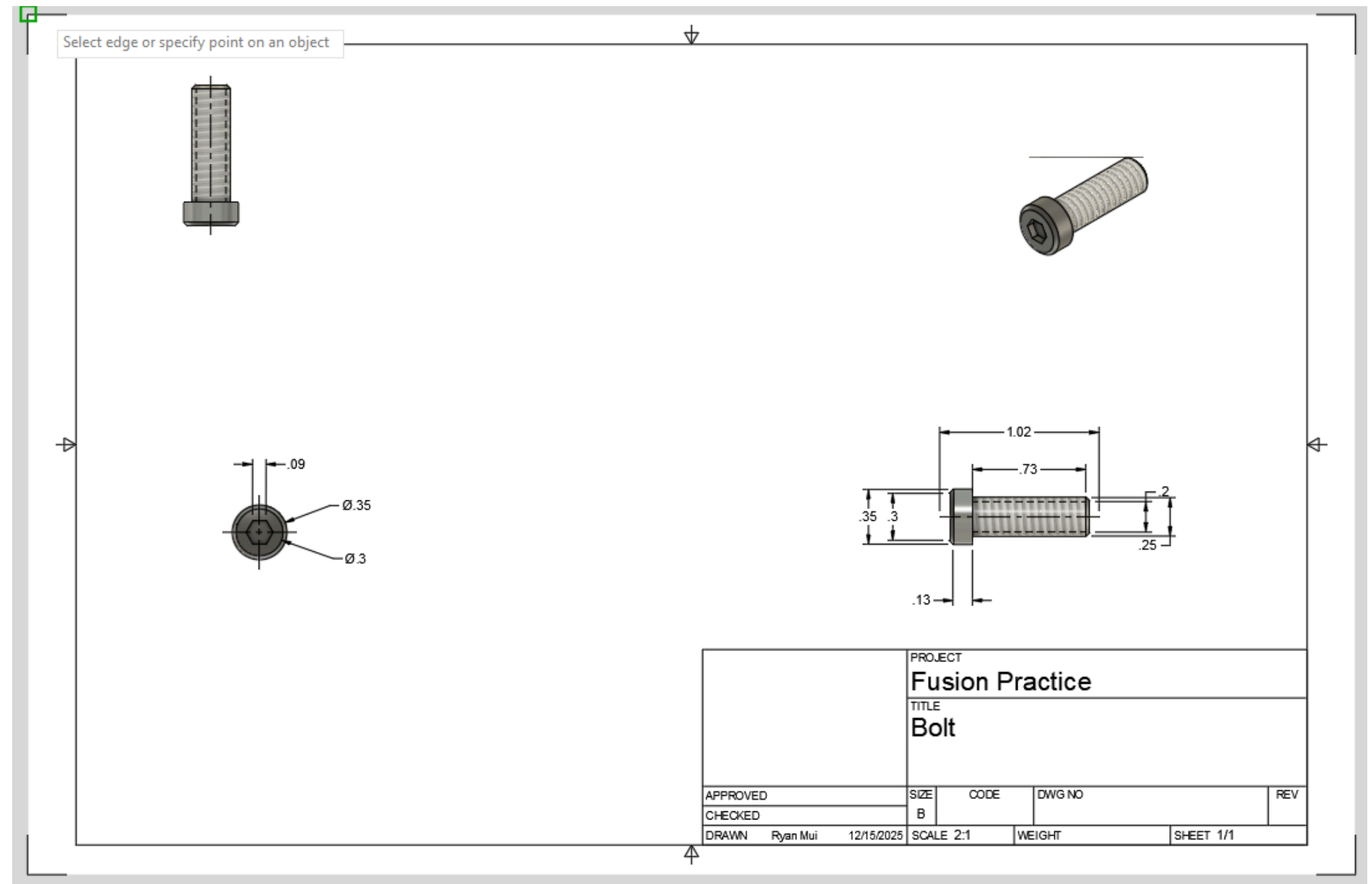
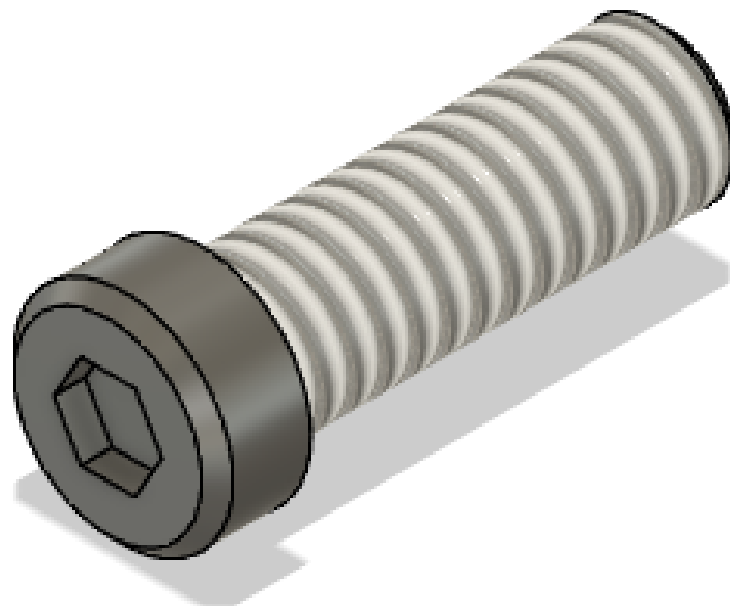
This Project Teaches:

1. Offset tool
2. Thread tool
3. Polygon tool (hex head)
4. Chamfering for manufacturability
5. Designing mating parts



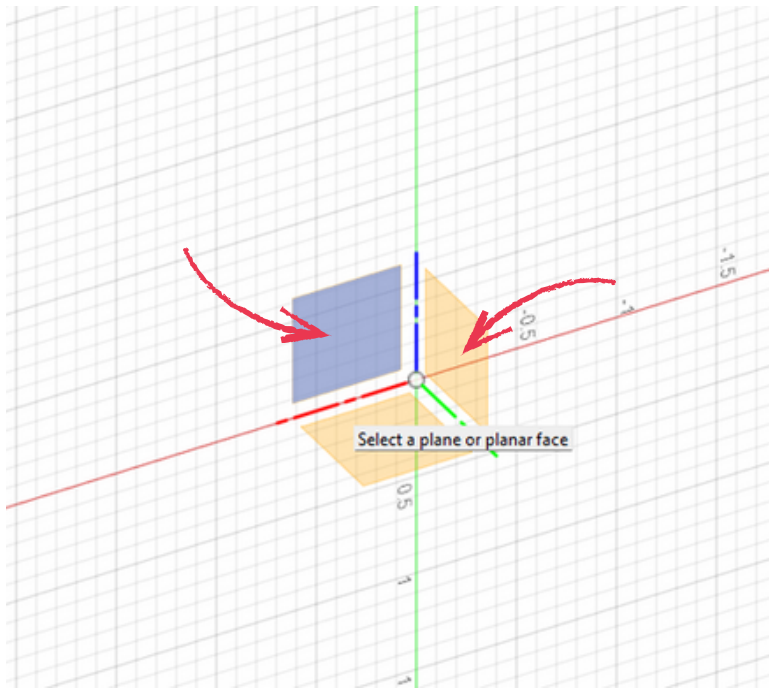


Bolt





Bolt Head

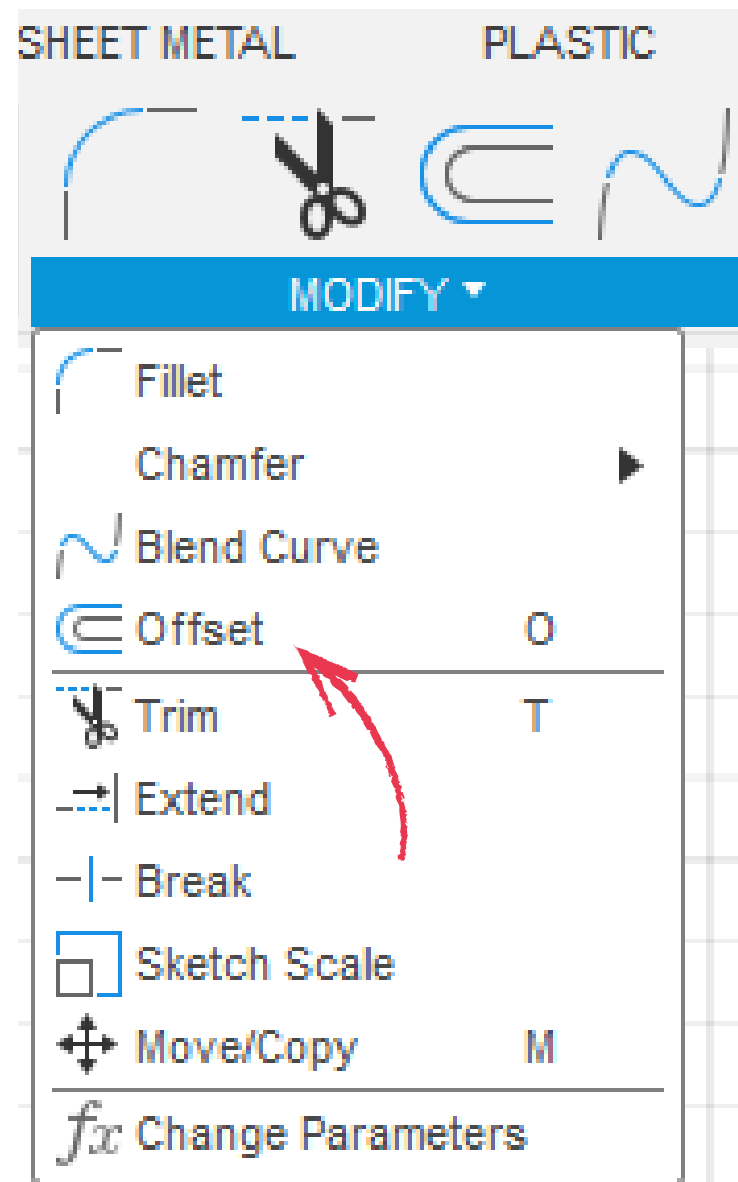


Let's make the bolt head really quick. Open up another new sketch and make a component. Like with the flower, select either the XZ or YZ axis and make a circle with a diameter of .3 in. Finish the sketch, and extrude the circle 0.15 in.

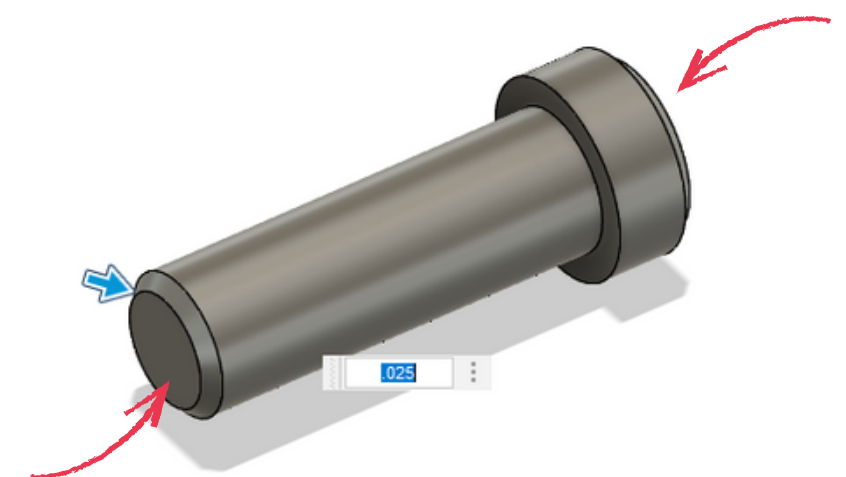
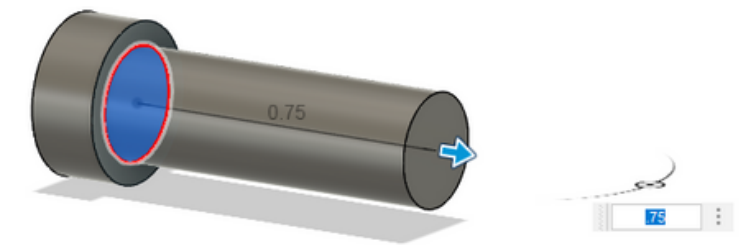
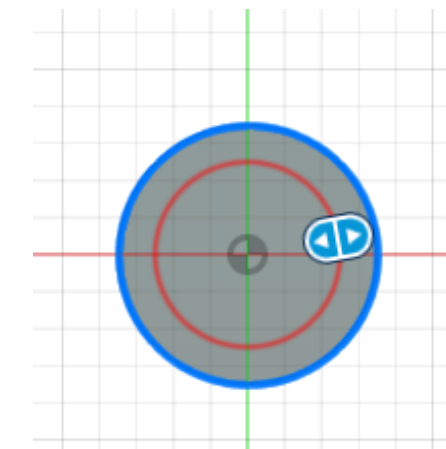




Offset Tool

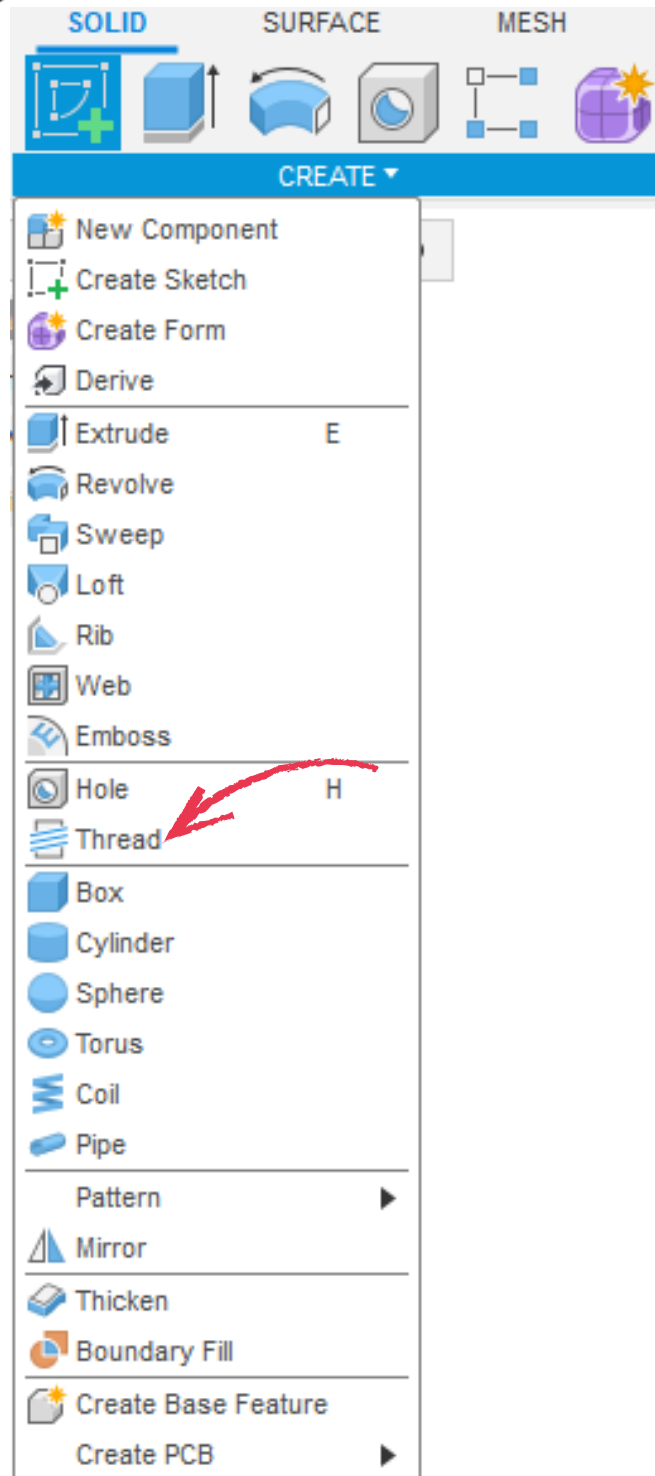


We're going to make the body of the bolt now. Start by making a sketch on the back face of the bolt head. Select the offset tool under the modify drop-down menu or shortcut O. Make sure to highlight the circle and enter an offset of 0.05 in. Make sure the offset is going inward as we want to make an inner circle. Finish the sketch and extrude the offset circle 0.75 in. Lastly, use the 3d chamfer tool on the ends of the bolt and add a chamfer of .25 in.

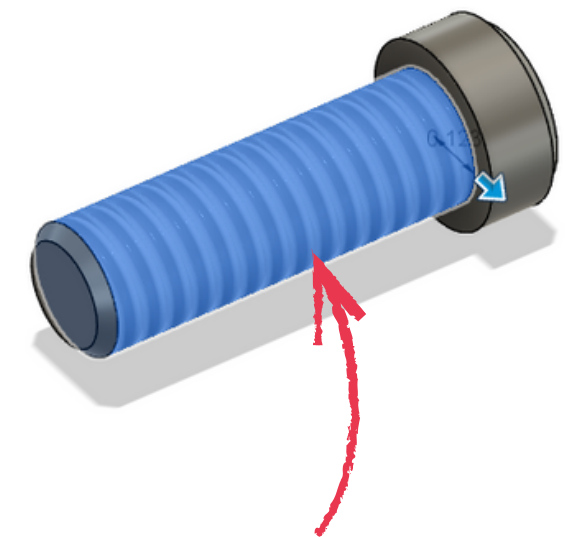




Thread Tool

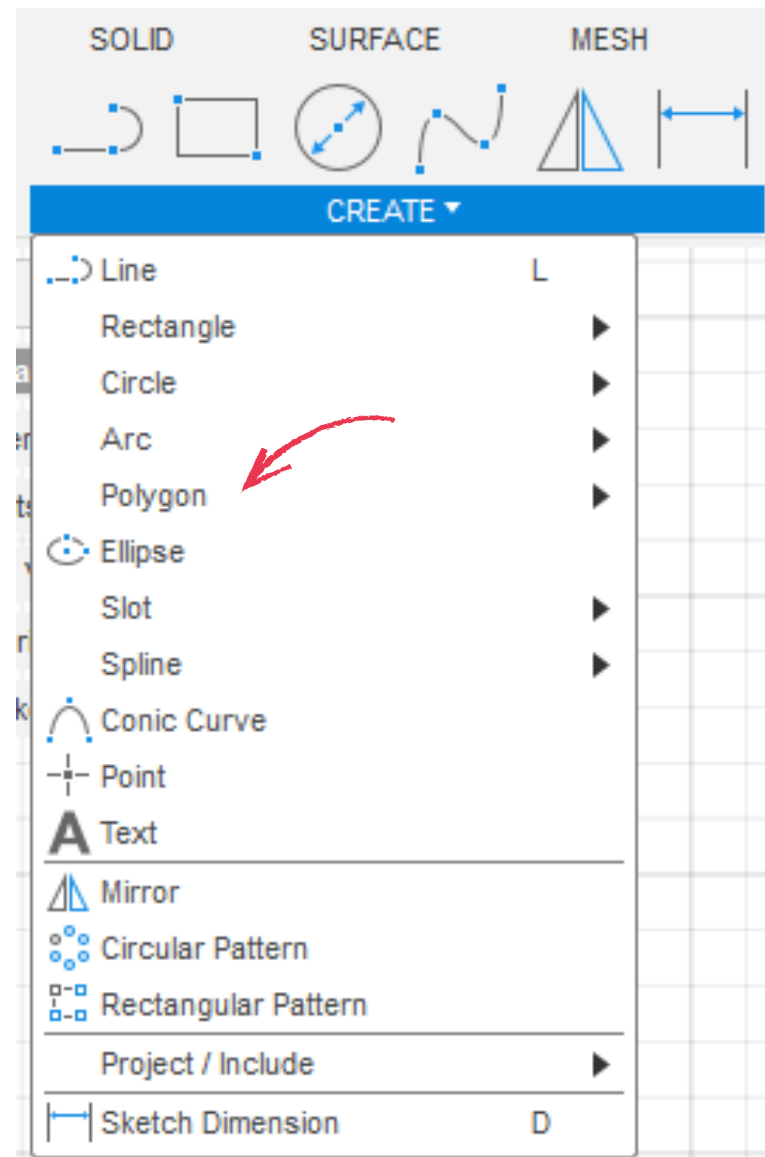


Let's add a thread to give our bolt the grooves that a real bolt would have. Under the create drop-down menu, select the thread tool and select the body of the screw. We'll keep it simple and stick to the basic thread options. Hit enter to lock in the thread.

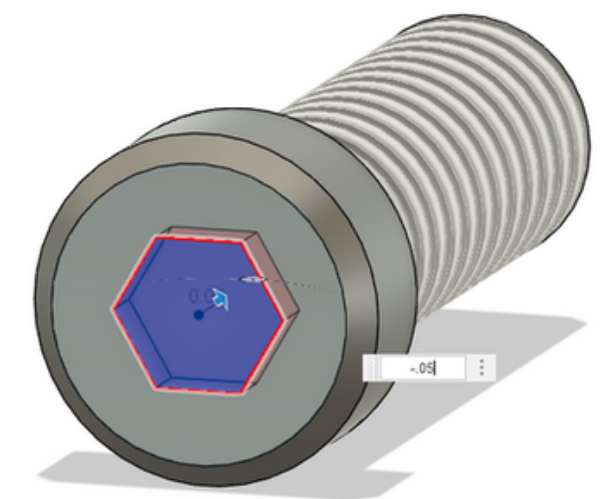
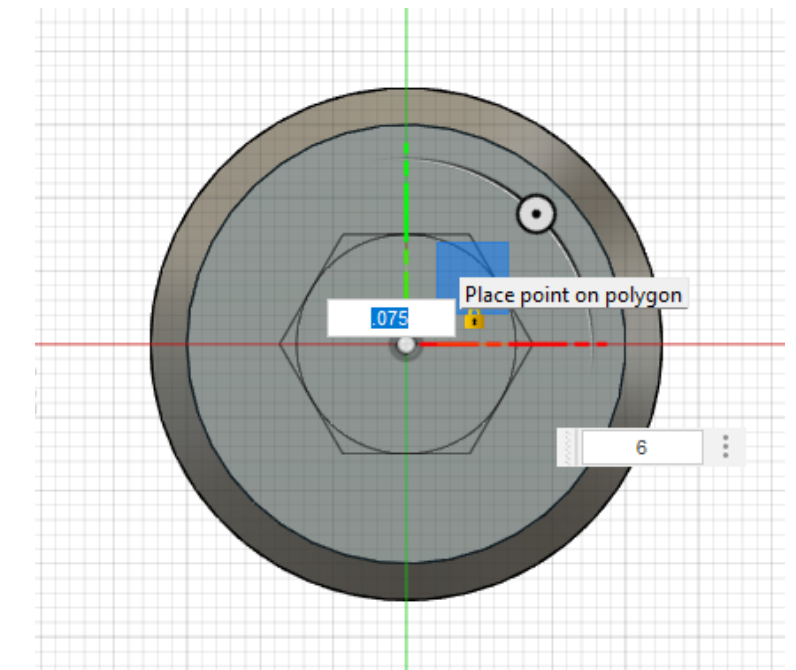




Polygon Tool



Readjust your camera and make a new sketch on the head of the bolt. Select the polygon tool and use the circumscribed polygon. Click at the origin, and a polygon will appear. The dimension box in the center displays the diameter, while the other box displays the number of sides. We'll keep the polygon as a hexagon; however, let's enter the diameter as .75 in. Lastly, finish the sketch and extrude the polygon 0.05 inwards. We're now done making the bolt. Make sure to save your model.



Project 5: Final Assembly!

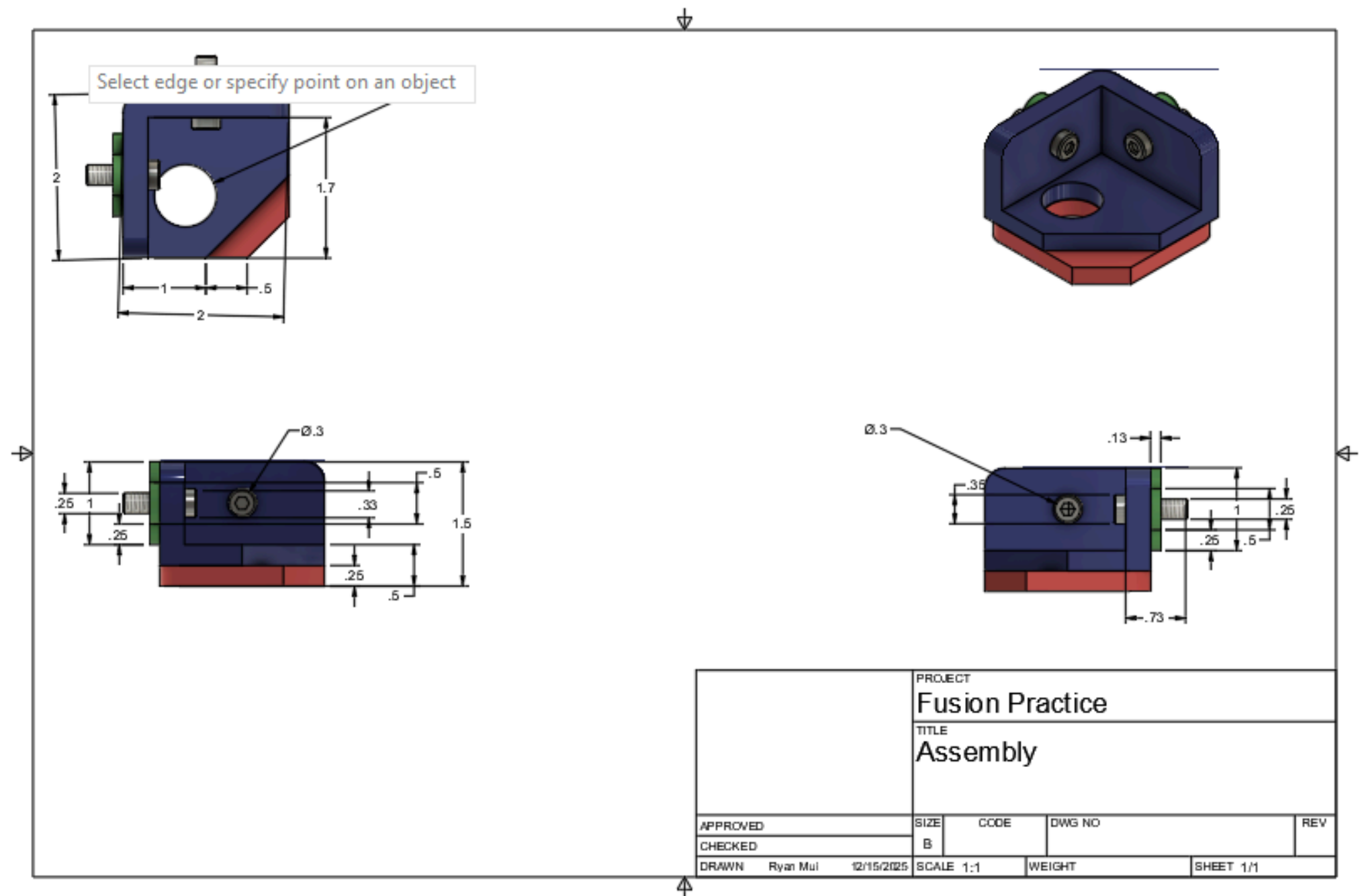
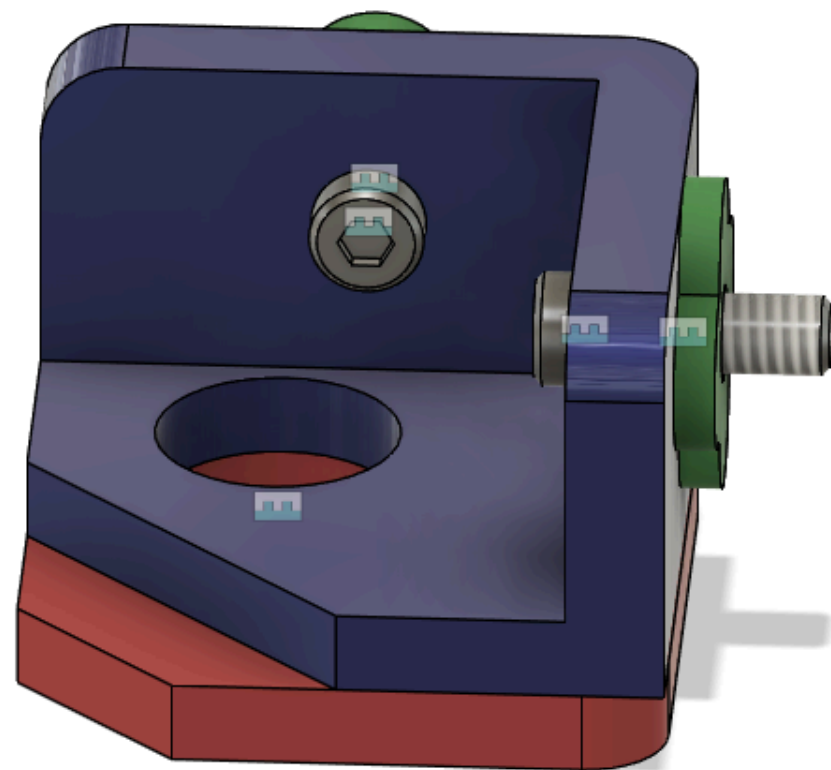
This Project Teaches:

1. Importing components
into a master design
2. Moving and rotating
parts in 3D space
3. Using rigid joints



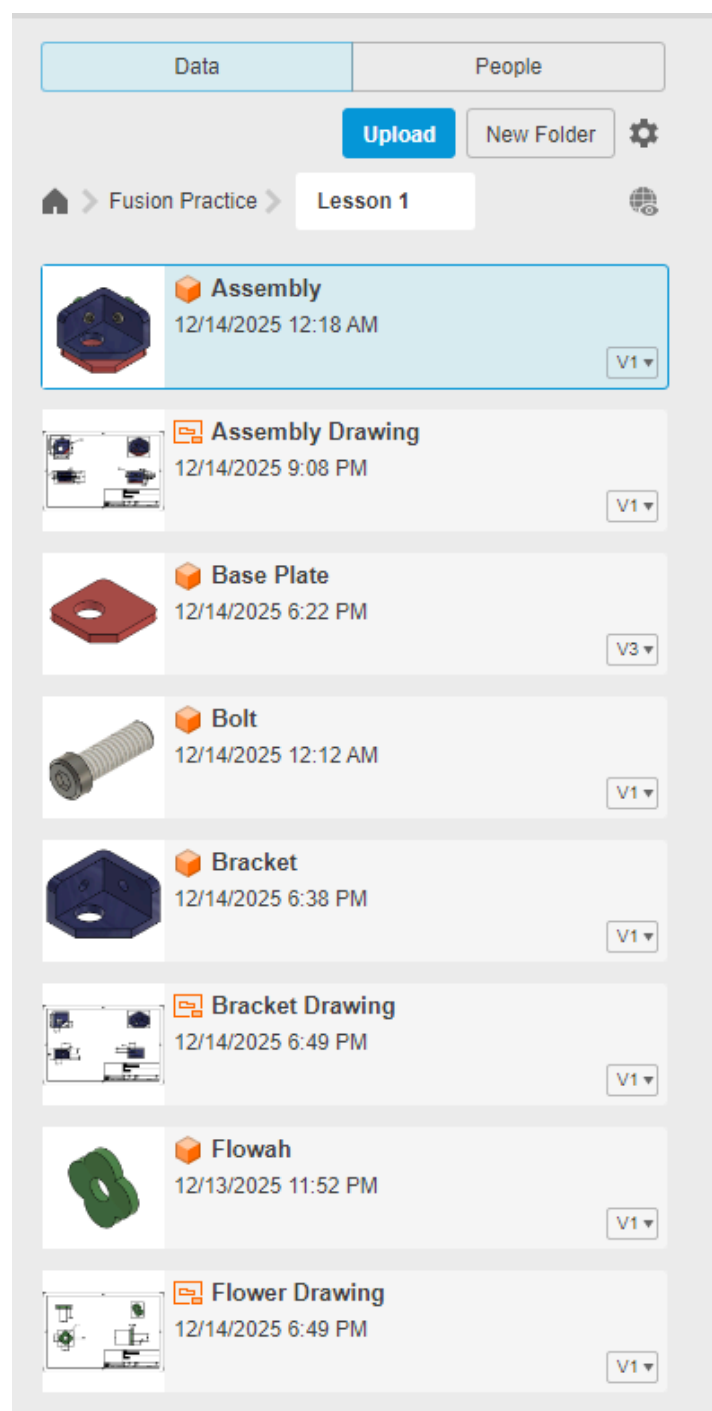


Final Assembly

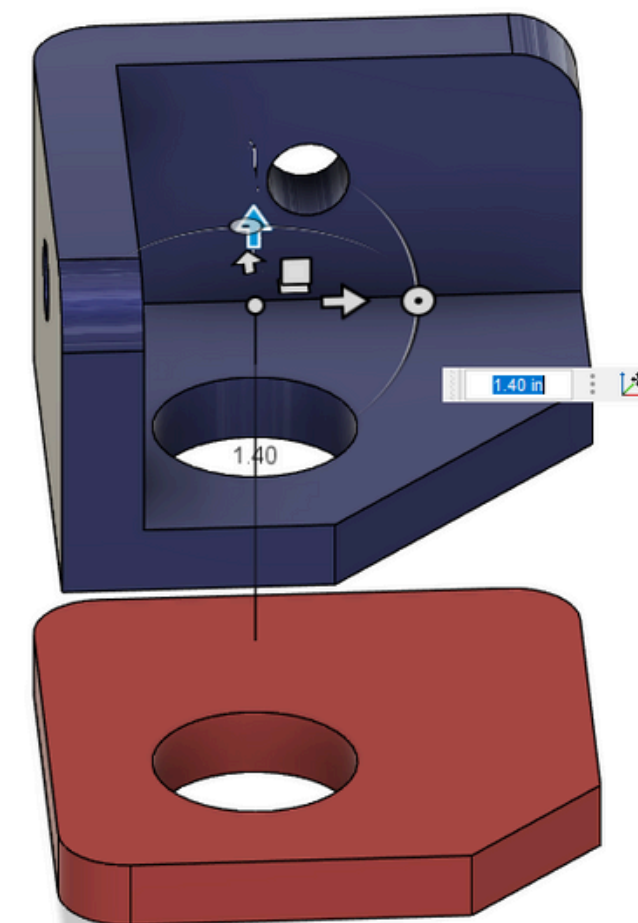




Adding Models



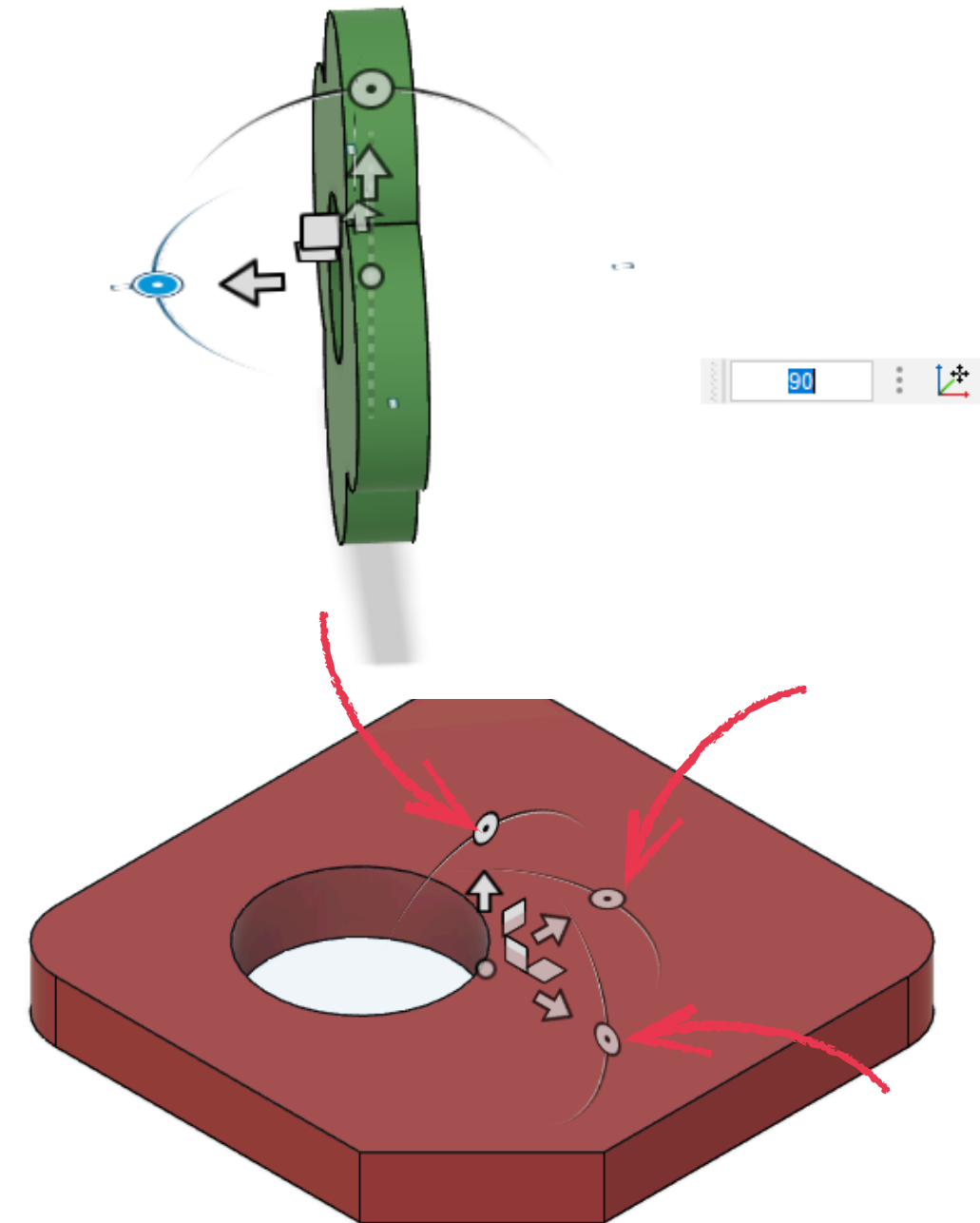
We're finally done with all our individual components, so let's combine them all now. Make a new model, but don't make a new component. Let's start with adding our base plate. Hold down the left click on the base plate and drag anywhere into our design. It should place it in the center of our grid. Let's do the same with the bracket. The bracket and base plate will overlap, so pull on the upward arrow to separate the two components. We don't need a specified distance, so just pull the bracket high enough so you can easily access its faces.





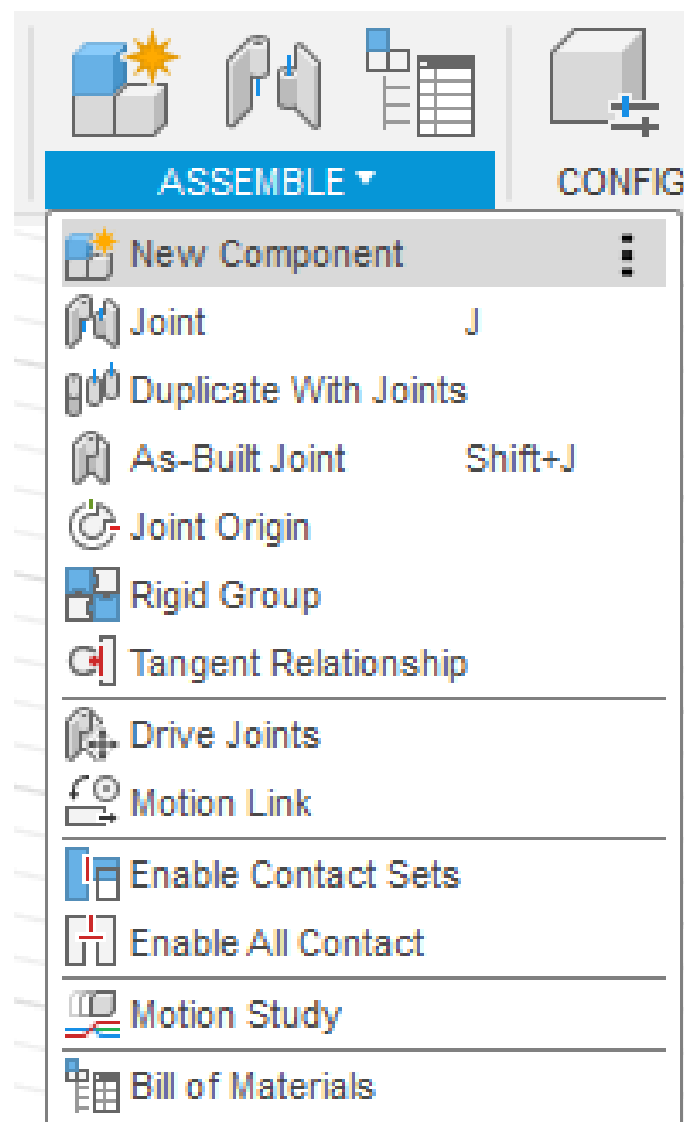
Important Note

Depending on what axis you selected or in what planes you decided to make your sketches/models, they may not line up at first with the rest of your model. When moving your components to a new design, you're given the ability to rotate your components to your desired position with the rotator icons. The three rotator icons allow you to rotate your component along all three planes.

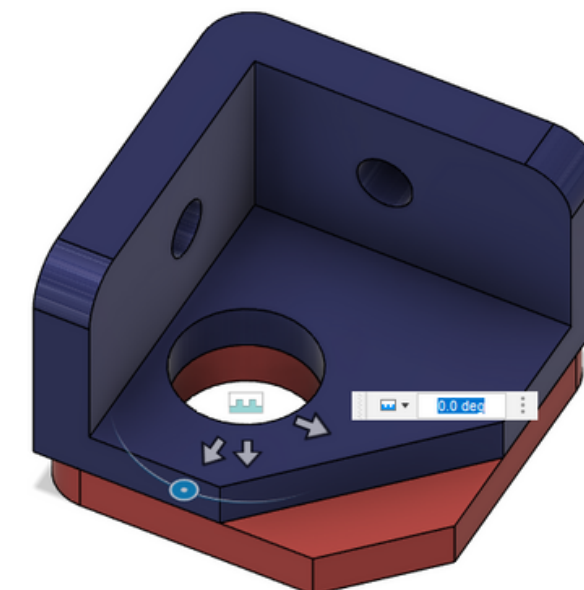
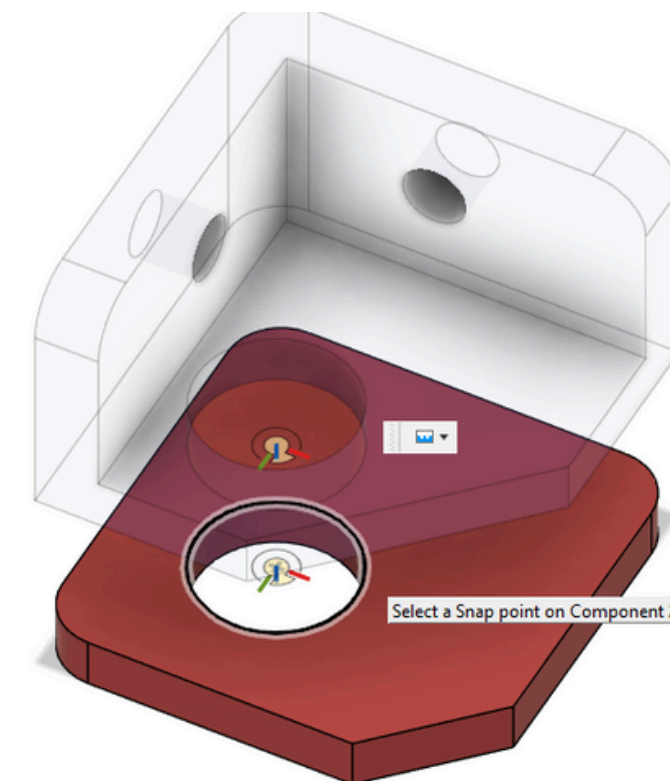




Jointing

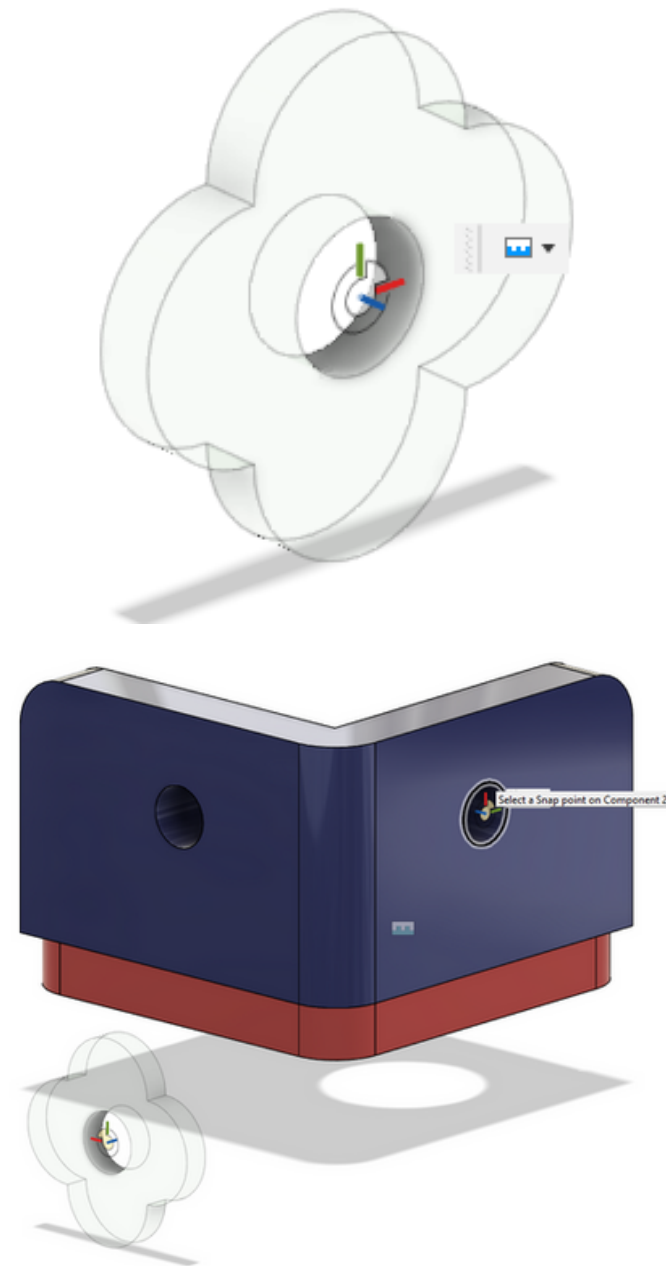


We're ready to joint the bracket to the base plate. Under the assemble drop-down menu, select the joint tool or shortcut J. Joints work by connecting and linking selected faces together. Since we want to place the bracket on top of the base plate, we'll first select the bottom hole outline of the bracket and then the top hole outline of the baseplate. This will link the two components together, as shown by the 0 degrees in between them. The small popup can change the type of joint; however, we'll just stick to rigid joints.

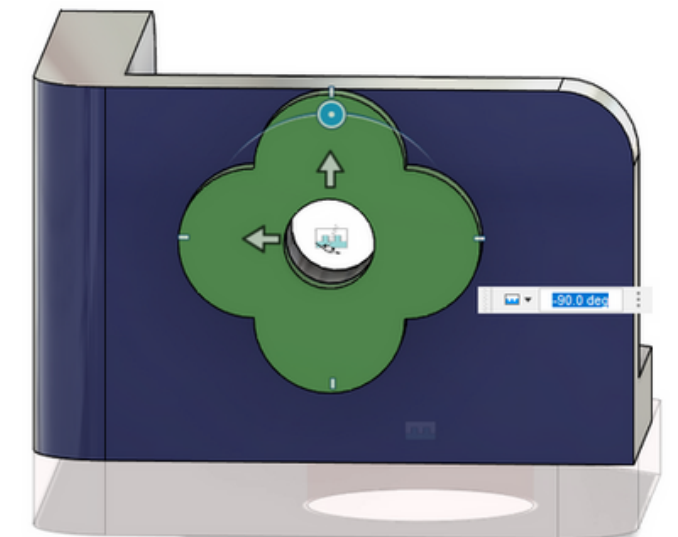




Jointing the Flowers



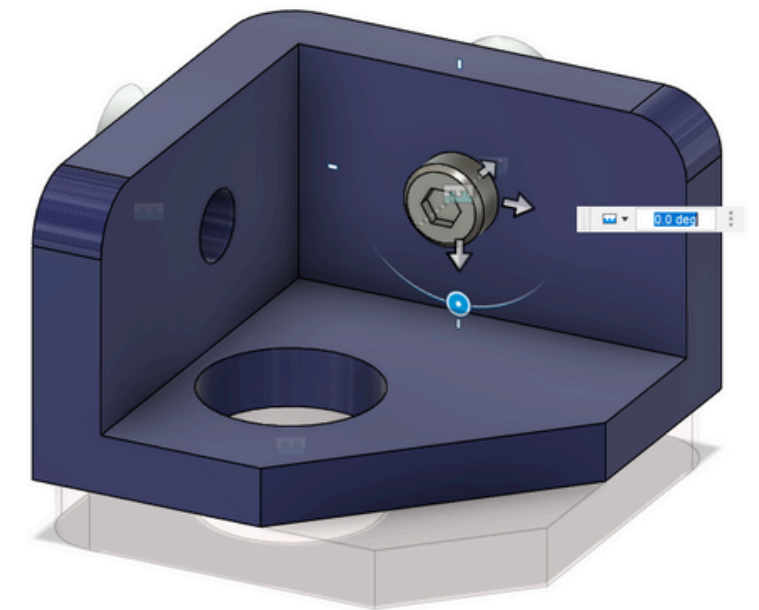
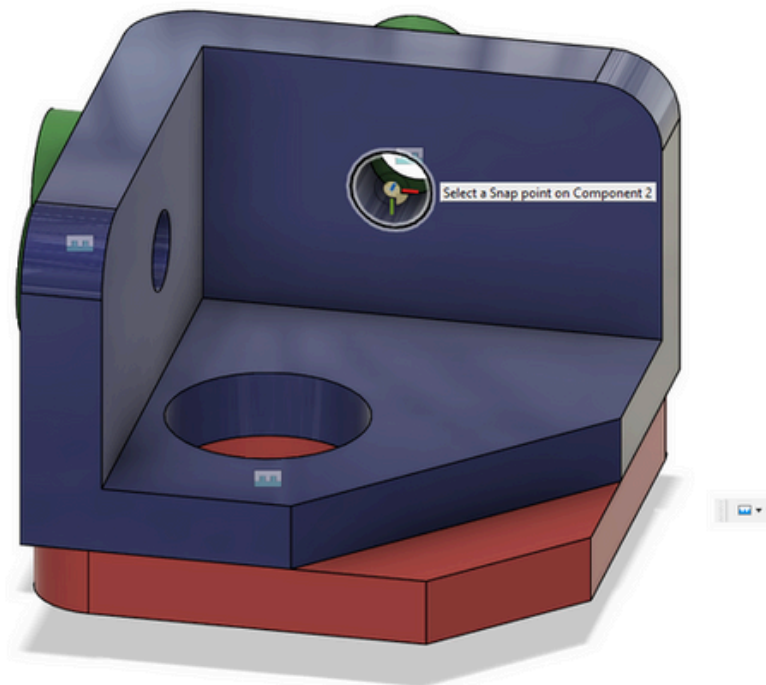
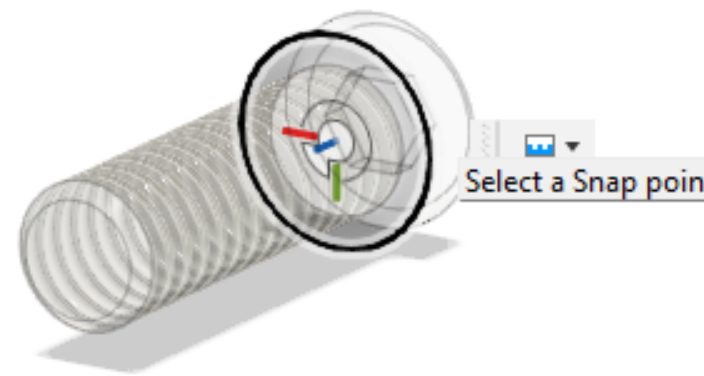
To joint the flowers to the bracket, it works the exact same. First select the center circle and joint them to the back circles on the bracket. If the flower becomes imbedded in the bracket, undo your joint and try jointing with the other side of the flower. I recommend rotating the flowers before jointing so they're parallel with your selected side of the bracket. Once you jointed one flower, drag in another copy and repeat the process.





Jointing the Bolts

Lastly, let's joint the bolts to the rest of the model. If we think about how the bolt would be inserted, we would want the bottom of the bolt head to be touching the inner circle. Select the face as shown in the left picture and join it to one of the inner circles on the bracket. Once completed, drag another bolt and do the same. We're now done with the model :D



Tolerance

This Project Teaches:

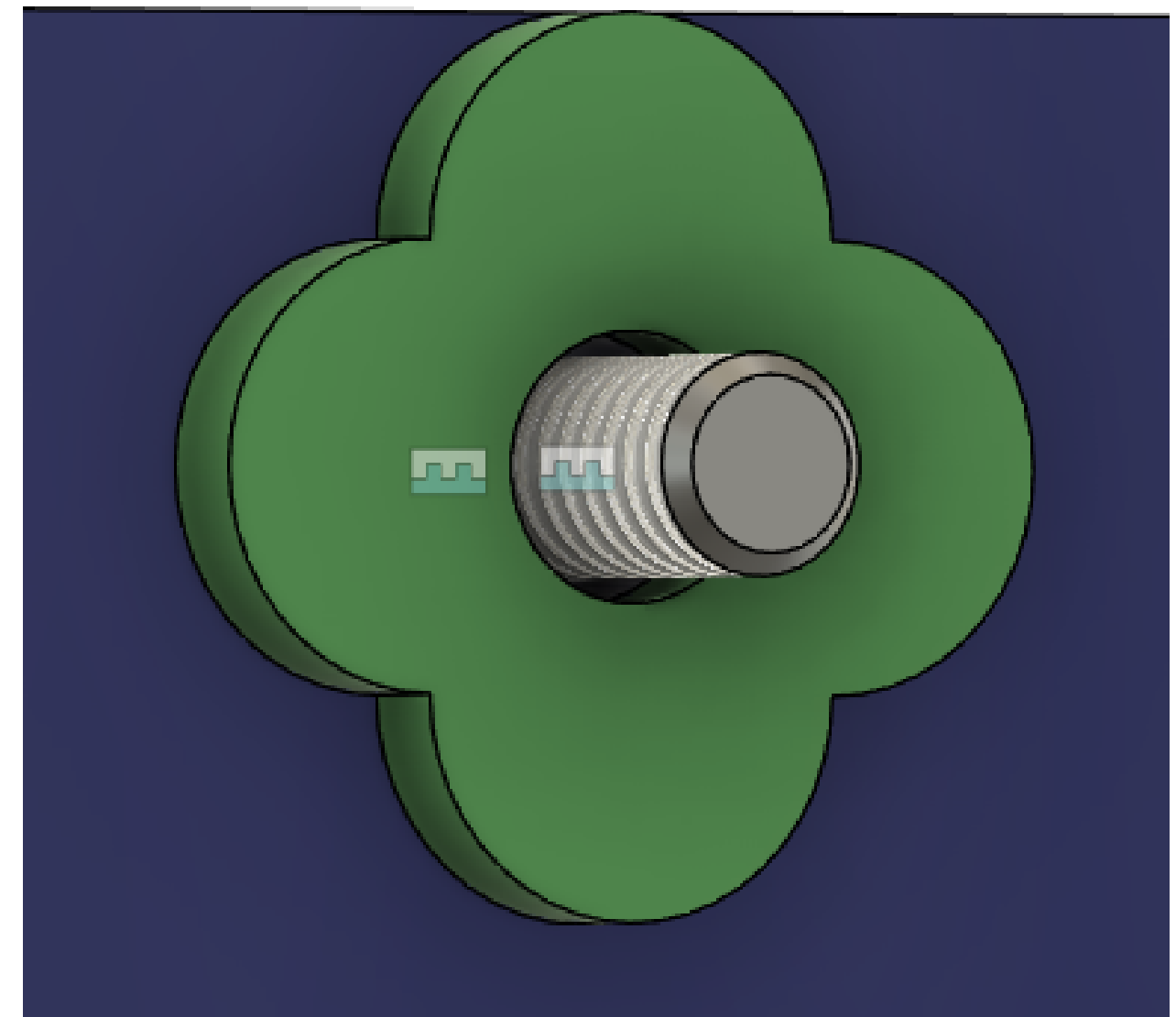
1. What tolerance is and why it matters
2. Types of tolerances
3. How tolerance affects assembly and usability





Intro to Tolerance

Notice how there seems to be extra room in-between the bolt and the holes? This wasn't done by accident. In engineering, this is known as accounting for tolerance.

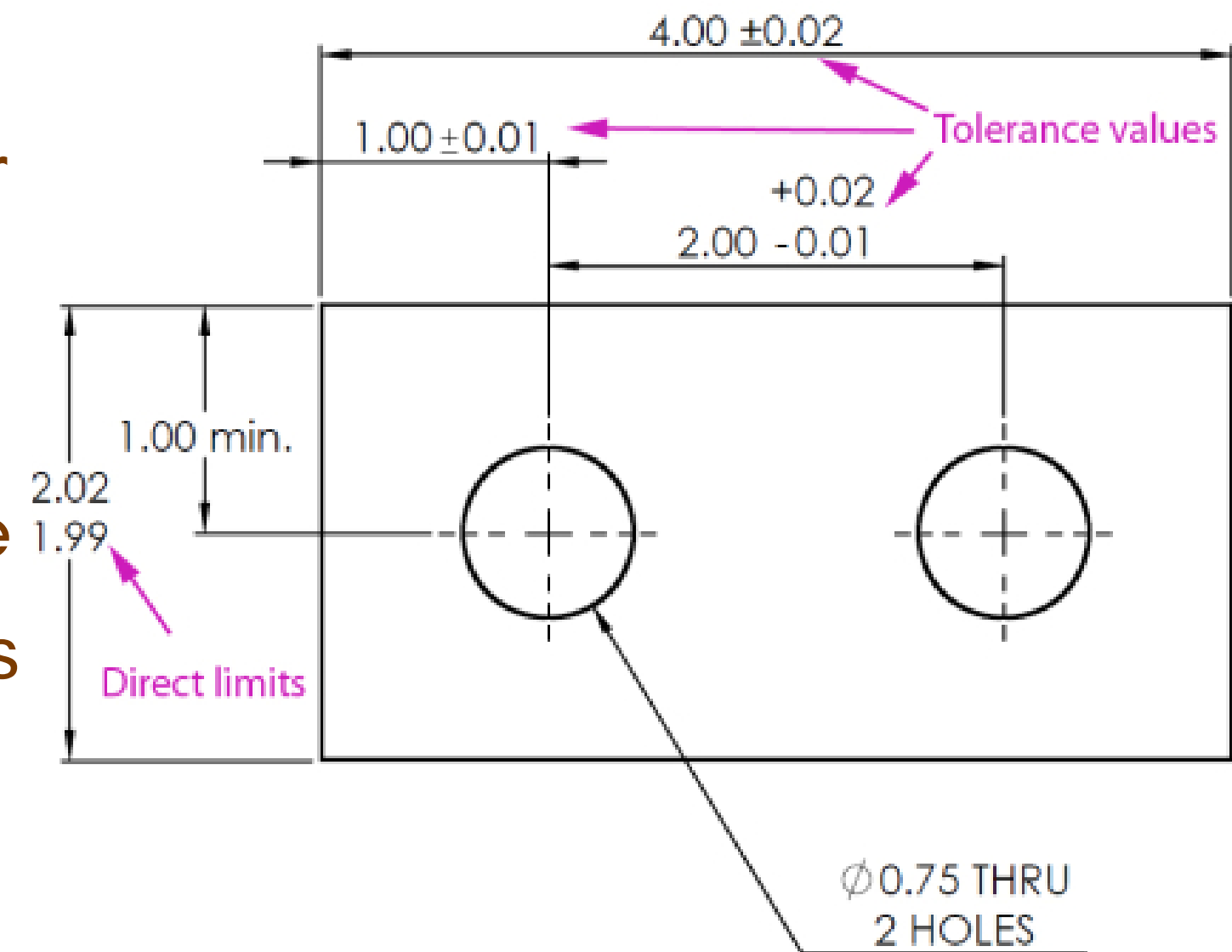




Definition of Tolerance

Tolerance is the allowable limit of variation in a physical measurement or characteristic of a part.

Because no manufactured part can be made perfectly, tolerance tells builders how much deviation is acceptable while still allowing the part to function as intended.





Unilateral vs Bilateral Tolerance

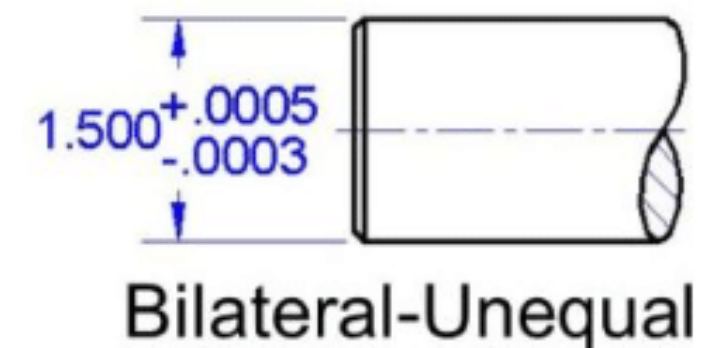
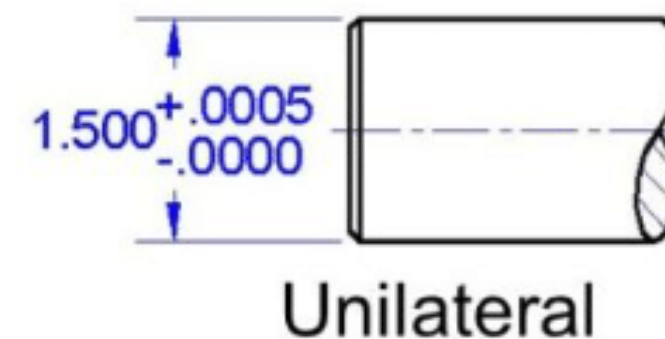
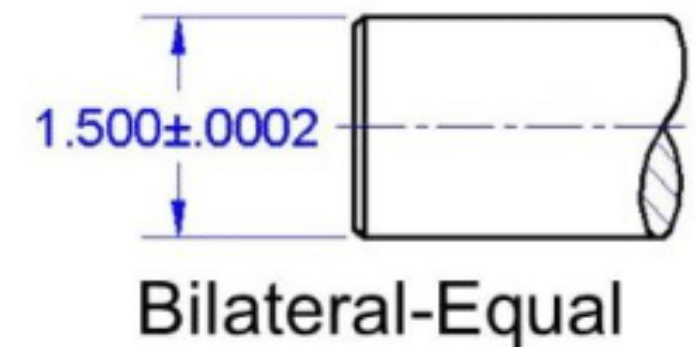
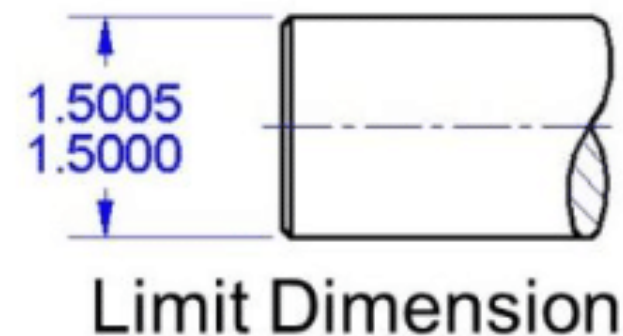
Unilateral tolerance varies in only one direction.

They can vary in either the positive or negative direction, but not both.

Bilateral tolerance varies in both directions.

Bilateral tolerance is usually indicated by a $\pm$ sign; however, unequal bilateral tolerances can exist, as shown on the right. Limit dimensions specify the maximum and minimum allowable sizes directly (e.g., 1.98–2.02 in), defining the full acceptable range.

It's important to adhere to specified tolerances when designing models to ensure functionality.





Different Types of Tolerance Fits

1

A **clearance fit** always leaves space between the shaft and the hole, even at the tightest tolerance limits. They're usually used when low friction is needed, or if easy assembly/disassembly is required.

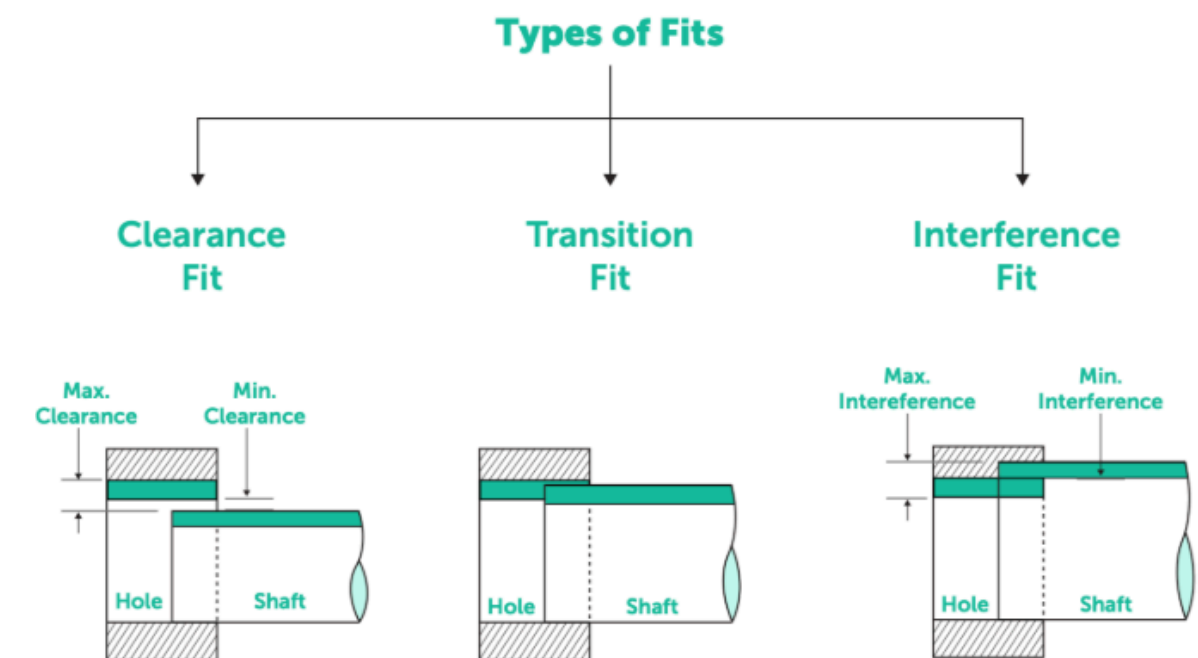
2

An **interference fit** means the shaft is larger than the hole at all tolerance limits. They are usually used for creating tight or permanent joints.

3

A **transition fit** can result in either a small clearance or small interference, depending on the tolerance limit. They are often used for creating accurate fits when components need to align precisely.

From our earlier model, the bolt was made to be a clearance fit so it could be easily removed.





Conclusion

You are now ready to design your own custom CAD enclosure for a project! This guide was designed to prepare you with starter Fusion 360 knowledge in order to add holes, screws, and other features when needed.

Have fun, and good luck!



ENCLOSURE

Design a custom enclosure for your project, get a 3D print of it shipped to you, and win other prizes!

